



The IMAP-Lo Instrument

N.A. Schwadron^{1,2} · M.M. Shen² · C. Amaya³ · S. Begley⁴ · J. Bower¹ · J. Brody³ · M. Bzowski⁵ · G. Cardarelli¹ · E.R. Christian⁶ · G. Clark⁴ · C. Collura⁴ · C. Cook⁴ · S. Cortinas³ · R. Crosier³ · M. Cully⁴ · A. De Los Santos³ · J. Dominico⁷ · N. Dutton⁴ · S. Ellis¹ · K. Fairchild¹ · J. Ford³ · C. Frost¹ · H.O. Funsten⁸ · S.A. Fuselier^{3,9} · A. Galli¹⁰ · M. Gargano¹⁰ · C.E. Garza³ · J. Gasser³ · M. Gkioulidou⁴ · M. Granoff¹ · N. Hall¹ · J. Hanley³ · D. Heirtzler¹ · C. Hersman⁴ · E. Hertzberg³ · S. Homan³ · I. Householder¹ · S. Hoyt¹¹ · H. Islam¹ · P. Janzen¹² · T. Jones¹ · N. Kerman¹¹ · B. King¹ · E. Klages⁷ · M.A. Kubiak⁵ · G. Lafferty¹¹ · W. Lam⁴ · C. Lomax⁴ · S. Longworth¹³ · K. Mathews⁴ · D.J. McComas² · C. McLeod¹⁴ · E. Möbius¹ · K. Nelson⁴ · A. Neto de Mattos¹³ · J. Niehof¹ · C. Nunez³ · D. Piazza¹⁰ · S. Pope³ · F. Rahmanifard¹ · D. Reisenfeld⁸ · J. Reiter¹¹ · C. Reno² · S. Rizo-Patron³ · H. Rodriguez³ · J. Rushforth¹ · C. Schiferl³ · C. Schlemm⁴ · M. Segler³ · J.M. Sokół³ · N. Souitai¹³ · D. Suffern⁷ · P. Swaczyna⁵ · M. Tapley³ · K. Taylor³ · J. Teifert² · W. Toczynski³ · P. Twombly¹ · S. Vogler¹ · C. Weaver¹ · A. Wester¹³ · M. Wieder⁴ · E. Wilber¹ · P. Wilson³ · R. Winslow¹ · P. Wurz¹⁰

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Abstract

The IMAP-Lo instrument is a large geometric factor, single pixel camera that measures energetic neutral atoms (ENAs) and interstellar neutral atoms (ISNs) over the low-energy portions of the ENA spectrum from 0.01 to 1 keV. It derives heritage from IBEX-Lo and shares a design philosophy with IBEX-Hi and IMAP-Hi. IMAP-Lo eliminates ion, electron, and ultraviolet photon background sources, and greatly reduces internal backgrounds compared to the heritage IBEX-Lo sensor. IMAP-Lo is mounted on a pivot platform for bore-sight articulation over a range of elongations ($\sim 60^\circ - 180^\circ$) from the spin-axis to enable tracking the interstellar flow. The instrument is divided into seven subsystems: the pivot platform articulates the instrument; the entry subsystem eliminates charged particles and determines the field-of-view; the star sensor provides pointing determination; the conversion subsystem converts neutrals to ions; the electrostatic analyzer determines the energy passband; the time-of-flight subsystem provides species separation (H, D, He, O, and Ne); and the electronics provides data and command interfaces, and supplies instrument high voltages. Measurements substantially reduce the ISN flow parameter uncertainties, mitigate systematic uncertainties, and provide accurate interstellar Ne/O and D/H ratios. The increased geometric factor and lower internal backgrounds of IMAP-Lo, coupled with the ability to pivot the boresight open an enormous array of new scientific opportunities that contribute to IMAP's quantum leap in understanding the composition and properties of the local interstellar medium, the evolution of the heliospheric boundaries, and the physics of interstellar-heliospheric interactions.

Keywords Heliosphere · Neutral atoms · Interstellar medium · Local interstellar medium

Extended author information available on the last page of the article

1 Introduction

The Interstellar Mapping and Acceleration Probe (IMAP) mission investigates two fundamental and interrelated questions at the heart of Heliophysics today. These questions are: (1) How are energetic particles accelerated in interplanetary space, and (2) How does the solar wind interact with the interstellar medium? These questions are investigated by taking groundbreaking new measurements to achieve the following objectives, as described in the IMAP mission overview (McComas et al. 2018, 2025):

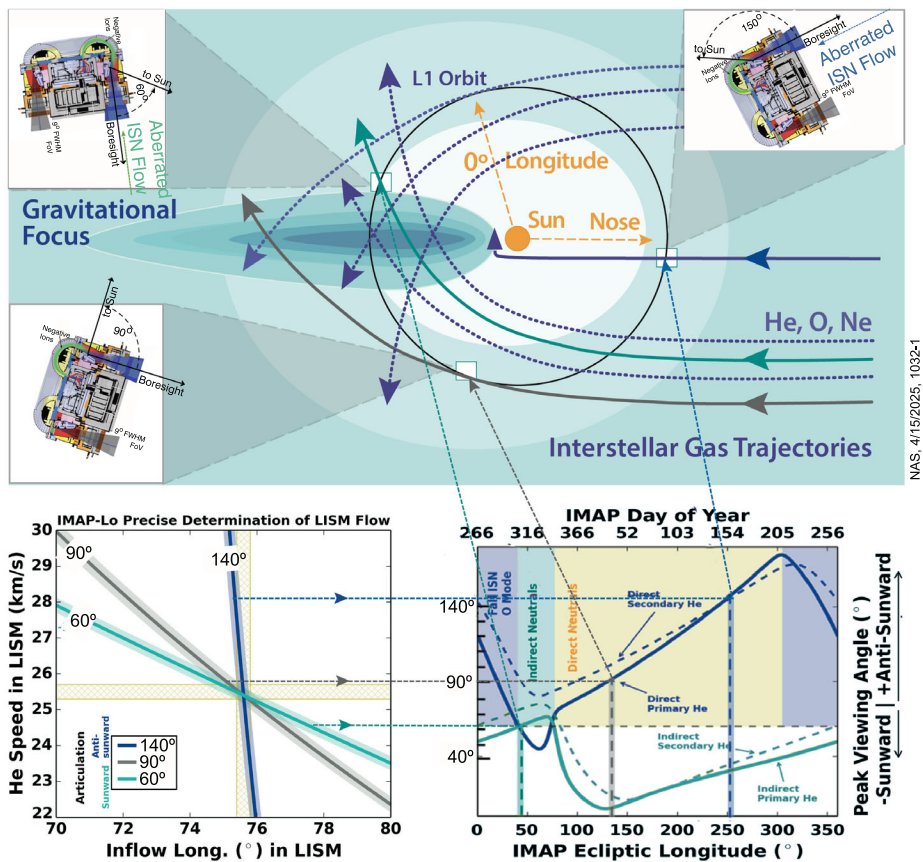
- 1 Improve understanding of the composition and properties of the local interstellar medium (LISM);
- 2 Advance understanding of the temporal and spatial evolution of the boundary region in which the solar wind and the interstellar medium interact;
- 3 Identify and advance the understanding of processes related to the interactions of the magnetic field of the Sun and the LISM;
- 4 Identify and advance understanding of particle injection and acceleration processes near the Sun, in the heliosphere and heliosheath.

The objectives are achieved using a combination of neutral atom measurements, energetic neutral atom (ENA) maps, dust measurements, observations of resonant backscatter glow of hydrogen and helium, and *in situ* solar wind, magnetic field, pickup ion, and energetic particle measurements. The ENA imaging provided by the IMAP-Ultra, IMAP-Hi, and IMAP-Lo instruments are critical for imaging the interstellar boundaries and the IBEX ribbon. The IMAP-Ultra, IMAP-Hi, and IMAP-Lo science objectives (see McComas et al. 2025; Reisenfeld et al. 2026) are related as they span the ENA spectrum from low energies 10 eV to 100's of keV to identify the processes that control the complex neutral and ionized plasma physics in the heliosheath. Cross-calibration (X-Cal) of IMAP with IBEX will enable fundamental discoveries regarding the time variations of the heliosphere and extend the IBEX dataset into the IMAP era over multiple solar cycles.

The development of IMAP-Lo has benefited from almost 20 years of development, test, and on-orbit operation of the IBEX-Lo sensor (e.g., Fuselier et al. 2009). The lessons from IBEX, and IBEX -Lo in particular, has had enormous impacts on refining the IMAP-Lo design, and in its development, and testing.

A major motivation for the IMAP-Lo instrument is to provide accurate measurements of interstellar flow parameters, which set the outer boundary conditions of the interstellar interaction (see McComas et al. 2025; Szalay et al. 2026). The velocity vector of the LISM relative to the Sun is a central quantity that regulates the interactions between global interstellar medium and the heliosphere (Szalay et al. 2026). The IBEX-Lo ISN measurements have a direct relationship between the ISN flow longitude and speed via the neutral atom trajectory equation (Lee et al. 2012). This relation indicates a large uncertainty in the longitude or speed (e.g., Schwadron et al. 2022; Swaczyna et al. 2022) because of the limited longitude range of the IBEX observations. An independent determination of the ISN longitude is needed to remove systematic uncertainties associated with the secondary ISN flow, while substantially tightening the determination of the flow vector. The pivot platform on IMAP-Lo provides the ability to track the interstellar flow over an extended longitude range and thereby increases the interstellar neutral atom coverage throughout the year.

The local interstellar magnetic field is important for regulating interstellar interactions between the solar wind and the nearby LISM (Szalay et al. 2026). The ISN flow vector and the interstellar magnetic field vector define the interstellar B-V plane, the natural symmetry



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Fig. 1 The IMAP-Lo pivot platform (PPM) enables ISN atom observations throughout much of the year for the precision determination of ISN flow properties, i.e., in tracking the independent parameter tubes of the ISN flow (Schwadron et al. 2022). The motion of IMAP-Lo is shown around the Sun at the L1 Lagrangian point. Interstellar atom trajectories are shown at different elongation angles. The trajectory in grey indicates the motion of interstellar He, O, and Ne that is measured near an elongation angle of 90° , as was done on IBEX-Lo. Trajectories are also shown (in blue) upstream of the Sun at a large elongation angle $\sim 140^\circ$ and (in green) downstream at an angle of $\sim 60^\circ$ where the ISN atoms pass through perihelion prior to reaching IMAP-Lo

plane of the outer heliosheath (e.g., Lallement et al. 2005; Schwadron et al. 2016). Secondary ISN populations created in the outer heliosheath flows from the direction within this B-V plane (e.g., Kubiak et al. 2016).

The LISM conditions derived by IBEX raise fundamental questions about the interstellar environment in the immediate vicinity of the Sun. Not only is the heliosphere close to the edge of the LIC, but it may be in a more variable environment near the interaction regions between multiple interstellar clouds (Szalay et al. 2026).

Figure 1 shows the motion of the S/C and IMAP-Lo around the Sun at the L1 Lagrangian point. Interstellar atom trajectories are shown at different elongation angles. The trajectory in grey shows the trajectory of interstellar He, O, and Ne that is measured near an elongation angle of 90° , as was done on IBEX-Lo. IBEX-Lo has a fixed orientation on the IBEX spacecraft, roughly 90° from the IBEX spin axis. In contrast, IMAP-Lo is mounted on a pivot

platform enabling observation of a range of elongation angles from $\sim 60^\circ$ to $\sim 180^\circ$. The capability to re-orient the instrument boresight throughout the year is critical for measuring the interstellar flow from different vantage points. Figure 1 also shows trajectories observed (in blue) upstream of the Sun at a large elongation angle $\sim 140^\circ$ and (in green) downstream at an angle of $\sim 60^\circ$ where the ISN atoms pass through perihelion prior to reaching IMAP-Lo.

Since 2009, Interstellar Boundary Explorer (IBEX) observations have been analyzed to return parameters associated with the bulk velocity vector and the temperature of ISN flow distributions. IBEX-Lo makes measurements from the same orientation, roughly 90° to the spin-axis, and over a similar range of ecliptic longitudes during each year's interstellar measurement campaign. These measurements are being made with the same instrument orientation and the same physical location during Earth's yearly motion around the Sun results in a degeneracy represented as a "4D parameter tube" since the four derived interstellar parameters (the two inflow direction angles, temperature, and speed) are not independent (Schwadron et al. 2022). The degenerate parameter tube is conventionally expressed in terms of the ISN speed, temperature, and inflow latitude direction each as functions of the inflow longitudinal direction. Breaking the degeneracy requires measurements over a much broader spatial region than provided on IBEX. The pivot platform enables changing the viewing elongation (Fig. 1) throughout the year. The first year of IMAP-Lo operations provides elongation angles from $\sim 75^\circ$ to $\sim 105^\circ$, which broadens the elongation angles measured to 30° . This will immediately provide a reduction in the ISN measurement degeneracy. In subsequent years, the measurement degeneracy will be further reduced as IMAP-Lo utilizes larger ranges of the pivot platform motion.

IMAP-Lo opens an enormous array of new scientific opportunities by increasing its geometric factor (G) relative to that of IBEX-Lo, reducing its internal instrument background using slight modifications of electrodes that can emit electrons and lowering the internal instrument temperature due to shadowing by the Sunshade, and enabling articulation of the instrument boresight using its placement on a Pivot Platform. These improvements in IMAP-Lo will be utilized to better resolve neutral atom populations from the interstellar medium and from those populations produced by charge-exchange within the heliosheath, to understand precise ISN flow properties, to decipher filtration in the heliosphere and the ionization degree of interstellar species, and to deduce ionization rates in the heliosphere. These measurements will provide a means to understand the complex thermodynamics and statistical physics of the local interstellar medium, and the implications of interstellar D/H and Ne/O for Big Bang cosmology and the interstellar volatile species inventory. Similarly, low energy ENAs measured by IMAP-Lo will reveal fundamental physics associated with the plasma interactions of the heliosheath that control the evolving boundaries of our heliosphere, the IBEX ribbon, and the neutral-ionized plasma in the nearby interstellar medium. The remarkable scientific potential of IMAP-Lo in the context of IMAP is detailed by McComas et al. (2025), Szalay et al. (2026) and Reisenfeld et al. (2026).

This paper details the IMAP-Lo instrument. Section 2 summarizes science requirements for IMAP-Lo. The instrument is described in Sect. 3: The main subsystems include the pivot platform (PPM, Sect. 3.2), the entrance subsystem (Sect. 3.3), the star sensor (Sect. 3.4), the conversion subsystem (Sect. 3.5), the electrostatic analyzer (ESA, Sect. 3.6), and the Time-of-Flight (TOF) analyzer (Sect. 3.7). The key instrument electronics include the TOF board (Sect. 3.8), the TOF High-Voltage Power Supply (TOF HVPS, Sect. 3.9), the Bulk HVPS (Sect. 3.10), and the Integrated Common Electronics (ICE) E-Box (Sect. 3.11). Section 4 describes the IMAP-Lo engineering model (EM). Section 5 summarizes the IMAP-Lo flight model (FM) calibration (Sect. 5.1), cross calibration (Sect. 5.2), and the IMAP-Lo response model (Sect. 5.3). Section 6 describes IMAP-Lo operations (Sect. 6.1), and IMAP-Lo data products and algorithms (Sect. 6.2). A final summary is provided in Sect. 7.

2 Science Requirements

Heliospheric and interstellar neutral fluxes are low and potential background contributions are very high. Propagation times for neutrals from the vicinity of the heliosheath to the inner solar system range from years (for 1 keV ENAs) to decades (for ISN atoms). Only ~ 55 to 80% of the 1 keV ENAs from the heliosheath propagating into the S/C can survive to be detected at L1, and this survival probability drops significantly to ~ 10 to 55% at 100 eV. The other lost ENAs are ionized along their trajectory, predominantly from charge-exchange and photo-ionization. The majority of ionization loss from charge-exchange is localized within the last 10 AU of the ENAs trajectory to the S/C. If the ENA path goes through the high-density plasma driven by a large transient, such as a coronal mass ejection (CME), there can be significant extinction of the ENA signal.

The very low fluxes and long timescales drive the IMAP-Lo instrument design to a large geometric factor, single pixel camera. Full energy and angle images with appropriate time resolution are accumulated by reorienting the (spinning) IMAP spacecraft each day as described in the mission overview (McComas et al. 2025).

On IMAP-Lo, heliospheric neutrals are ionized inside the instrument and resulting ions are deflected away from their incident trajectory, thereby separating the signal from potential backgrounds such as UV. Triple coincidence measurements are used to combine high detection efficiency with high background rejection.

IMAP-Lo provides much larger collection power than IBEX-Lo, using increased geometric factors (by more than $4\times$) and its position at L1 to increase viewing time with lower backgrounds. There are three elements that promised increases in geometric factor at the time of IMAP-Lo's design (McComas et al. 2018): the larger $9^\circ \times 9^\circ$ field-of-view increases G by $1.9\times$, the removal of IBEX-Lo high-resolution sector increases G by $1.2\times$, and the high throughput mode with broadened energy acceptance, which increases G by $1.7\times$. The combination of these factors increases G by $\sim 3.9\times$ compared to IBEX-Lo.

Geometric factors derived from IMAP-Lo calibration are even larger than those estimated during design (the overall increase in G is $\sim 6\times$ compared to IBEX-Lo). These additional increases in G are due to increased efficiencies through the use of thinner carbon foils in the Time-of-Flight subsystem (see Sect. 3.7), increased throughput due to the removal of unneeded field-shaping grids behind the entrance system (see Sect. 3.3), and increased conversion efficiencies due to slightly enlarged conversion surfaces (Sect. 3.5).

The IMAP-Lo instrument is dual-use: it must be capable of measuring interstellar neutral atoms (H, D, He, O, Ne) throughout much of the year while also delivering high quality ENA maps of the heliosphere down to low energies (< 200 eV). The need to resolve the ENA energy distributions drives the mapping requirement of an energy resolution similar to that of IBEX-Lo, $\Delta E/E \sim 70\%$ (the Hi-resolution, HiRes mode), while providing a significantly increased geometric factor. The latter requirement drove the development of a Hi-throughput (HiThr) instrument mode with broadened $\Delta E/E \sim 100\%$.

Another factor in the development of IMAP-Lo was the need to make very accurate measurements of interstellar Ne/O and D/H ratios. Both measurements rely, in part, on the ability to accurately segregate interstellar neutral atom species within the IMAP-Lo TOF subsystem. The use of $1 \mu\text{g}/\text{cm}^2$ carbon foils (as compared to $2 \mu\text{g}/\text{cm}^2$ foils utilized by IBEX-Lo) reduces the energy loss and energy straggling of ions passing through foils, with the latter significantly increases the mass resolution of TOF peaks. The use of thinner foils has another significant advantage over IBEX-Lo is increasing the detection efficiencies through the time-of-flight system, thereby contributing to the increased geometric factor discussed previously.

Table 1 IMAP-Lo Science Requirements Summary

Requirement	Specification	Allocation
Detection, Separation H, D, He, Ne, O	$M/\Delta M \geq 4$	CS, TOF & Ion Optics
ISN Species Observations	60 d in 2 yr	PPM (60°-180°)
H ENAs	0.5-1 keV	Increased G ($\times 6$), CS & Ion Optics
Independent He observations	$> 30^\circ$ in ecliptic longitude	PPM (60°-180°), CS & Ion Optics

Absolute pointing knowledge of IMAP-Lo to $\leq 0.1^\circ$ accuracy is achieved by a collocated star sensor with the same boresight direction as the IMAP-Lo instrument within extremely tight mechanical tolerances. The design is identical to that of IBEX-Lo (Hond et al. 2012), except for the photomultiplier (PMT) and an edge filter for wavelengths in the red to reduce the sensitivity to the Milky Way and Zodiacal Light background.

The PPM orients the IMAP-Lo boresight up to 180° from the spin axis, which is more than adequate to follow the interstellar flow through more than 180 days of the year (see Fig. 1). This enables high counting statistics for Ne/O and D/H (Kubiak et al. 2023, 2024). The varied orientation of the IMAP-Lo instrument breaks the IBEX-Lo ISN measurement degeneracy (Fig. 1, bottom-left panel).

The IMAP spacecraft provides clear fields-of-view for both IMAP-Lo and the star sensor over pivot-angles from 61° to 180° . The cutout in the S/C deck allows for clear views at large pivot angles. The 60° position is the launch-lock position for the instrument, and the 90° position, similar to that of IBEX-Lo, is the nominal configuration used for mapping the heliosphere.

Table 1 provides a summary of mission level science requirements for IMAP-Lo. The ability to distinguish species (H, D, He, O, and Ne) depends on the TOF system with resolved TOF distributions ($M/\Delta M \geq 4$). This requirement applies to the converted and sputtered species (H, D, C, and O) detected in the TOF system. The PPM is required for detection of ISN species over 60 days in 2 years to track the ISN flow, and to make independent He observations of ISN parameter tubes separated by $> 30^\circ$ in longitude, see Fig. 1. Detection of H ENAs below 1 keV requires the CS and ion optics with a broad energy passband to achieve a large geometric factor.

3 IMAP-Lo Instrument Description

3.1 Introduction

IMAP-Lo is a single pixel camera for imaging energetic neutral atoms and measuring interstellar neutral atoms from the flow of interstellar neutral atoms. IMAP-Lo is based largely on the heritage design of IBEX-Lo (Fuselier et al. 2009). Unlike IBEX-Lo, IMAP-Lo includes a PPM, enabling the instrument to “follow” the interstellar flow throughout much of the year, and observe ENAs from various pivot angles. The main components of the instrument, the IMAP-Lo instrument, the PPM, and the E-Box are shown in Fig. 2.

IMAP-Lo subsystems are depicted in Fig. 3 with incident neutrals shown along the blue trajectories into the instrument, green negative ions exiting the conversion surface (CS), and blue ions progressing into the TOF system. The large annular entrance system collimates incident neutral atoms (blue trajectories), and the deflector system deflects out incident ions and electrons. After passing through the collimator, atoms strike the conversion surface (a

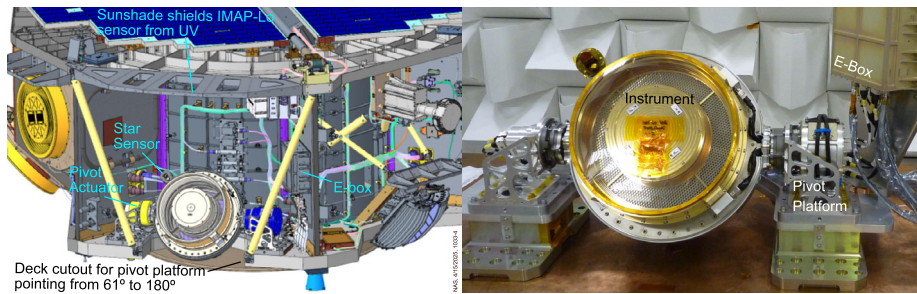
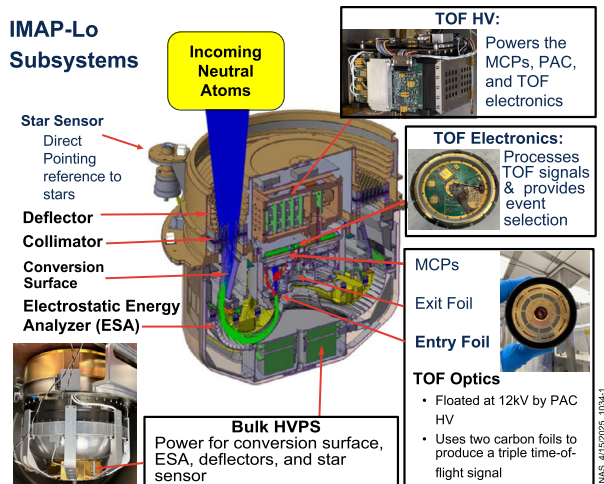


Fig. 2 (Left) The IMAP-Lo instrument on the spacecraft showing placement of E-Box, the actuator on the PPM, the sunshade and the star sensor; (Right) Components including the PPM, the instrument, and the E-Box shown during testing at JHU/APL

Fig. 3 IMAP-Lo cutaway drawing overlaid with SIMION ray-tracing (left side) of ENAs (blue), negative ions after conversion (green), neutrals and ions after passing the C foil(s) in the TOF section (blue), and secondary electrons emitted at the foils (red). The key subsystems are indicated in the figure: star sensor, deflector, collimator, CS, ESA, Bulk HVPS, TOF optics (including the entry foils, exit foils, and microchannel plates), the TOF electronics, and the TOF HVPS



diamond-like carbon, or DLC, surface serving as an electron donator) at a shallow $\sim 15^\circ$ angle. A fraction of the incident neutral atoms is converted into negative ions (green trajectories) either through conversion (the atom receives an electron from the conversion surface) or through sputtering of atoms and molecules (such as water molecules) off the conversion surface. The singly charged negative ions from the conversion surface are then accelerated into a curved (bundt-pan shaped) ESA, which filters ions based on their energy-per-charge. The ions then exit the ESA and are accelerated and electrostatically focused into the TOF subsystem, which is floated to ~ 12 kV. Table 2 summarizes the instrument parameters and resources.

IMAP-Lo consists of seven major subsystems: (1) the PPM, (2) the entrance subsystem, (3) the star sensor, (4) the CS subsystem, (5) the ESA, (6) the time-TOF subsystem, and (7) the E-Box. Four of these systems (the CS, ESA, TOF, and entrance subsystem) are mounted to the “optics deck”, the mechanical platform for ion optics and the mechanical mounting to the PPM. These seven subsystems are designed to operate as a unit, to maximize collection power while minimizing background.

Table 2 IMAP-Lo instrument parameters and resources

Resource or Parameter	Value or Range
Energy Range (eV)	5 - 1000
Energy Resolution ($\Delta E/E$, HiRes Mode)	0.7
Energy Resolution ($\Delta E/E$, HiThr Mode)	1.0
Mass Range (amu)	1 - 32
Mass Resolution ($M/\Delta M$)	≥ 4
Field-of-view	$9^\circ \times 9^\circ$
Pivot Range (angle to spin-axis)	$61^\circ - 180^\circ$
Geometric Factor ($\text{cm}^2 \text{ sr eV/eV}$), HiRes ($\sim 100 \text{ eV}$)	$1.1\text{e-}4$
Geometric Factor ($\text{cm}^2 \text{ sr eV/eV}$), HiThr ($\sim 100 \text{ eV}$)	$2.6\text{e-}4$
Total Mass (kg)	44.573
Instrument Mass (kg), with MLI excl. PPM	17.068
Ebox Mass (kg)	4.45
Volume on PPM with Harness	$65,750 \text{ cm}^3$
Volume Ebox	5300 cm^3
Power (W) without PPM	21.62
Power (W) with PPM	38.11
Telemetry (bps)	686

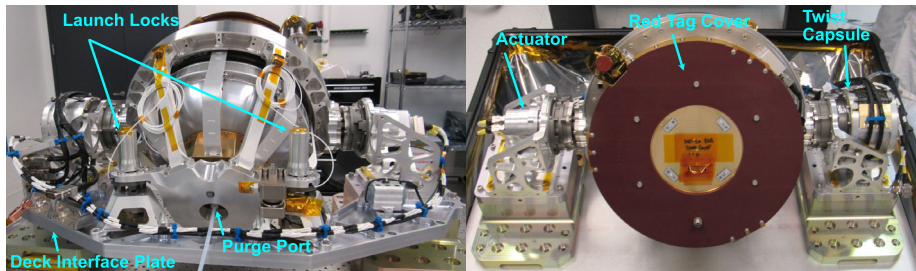


Fig. 4 Photographs of the assembly of IMAP-Lo at JHU/APL prior to environmental testing. Views are shown from the rear (left) and front (right) of the instrument with the PPM in its launch-locked position. The red-tag cover (right) blocks the aperture while the instrument is purged in a laboratory configuration using the purge port at the back of the bulk power supply. The HDRM launch locks are visible in the left image. The twist capsule and actuator are visible in the right image. The PPM mounts to the deck interface plate, which, in turn mounts to the spacecraft deck

3.2 Pivot Platform

The PPM (Fig. 4) enables IMAP-Lo to track the ISN flow over almost the entire year for vastly improved sampling of ISN populations compared to IBEX-Lo. With the PPM in the 90° pivot angle (instrument FOV at 90° to spin axis, identical to IBEX-Lo), IMAP-Lo views both heliographic poles every spin and creates a full sky ram map each year.

The interface between the PPM and spacecraft is provided by the Deck Interface Plate (Fig. 4). This interface includes features for precise alignment. The interface between the PPM and the IMAP-Lo instrument is provided by the Payload Ring, which includes dowel pins for alignment.

The PPM is electrically controlled from the pivot control card (PCC), which is mounted within the E-Box (Sect. 3.11). The electrical interfaces of the PPM include connectors that mate to the E-Box backplane, two that mate to the spacecraft power system, and three that mate to the IMAP-Lo instrument through the payload ring within the twist capsule.

IMAP-Lo is launched at a 60° pivot angle at a hard stop in its launch-lock position. The launch-lock is achieved with two electrically redundant Hold Down and Release Mechanisms (HDRMs) that restrain the pivoting mechanism via mechanical interfaces with mating kinematic elements to provide a robust load path during launch. In the launch-lock position, a purge connection through the bulk power supply is made near the back of the instrument. One of the first operations for IMAP-Lo during commissioning is to release the launch lock, after which the pivot platform is moved out of its launch-lock configuration.

3.3 Entrance Subsystem

The entrance subsystem comprises the sunshade, the deflection system, and the collimator. The IMAP S/C spin vector is set nominally 3.5°–4.5° to the right of Sun pointing in the direction of the aberrated solar wind. The sunshade extends from the top-deck and shields IMAP-Lo from solar UV, eliminates scattered sunlight, and solar wind from the entrance up to 12° from the spin direction. The 12° angle was chosen based on observed flow angle variation within the solar wind. Elimination of sunlight reduces UV flux into the instrument to interstellar levels (~ 0.8 kRayleighs maximum flux at Lyman Alpha wavelength).

The ion rejection design of IBEX-Lo was changed to a deflection design for IMAP-Lo (Fig. 5) to completely eliminate the “ion gun” effect (Wurz et al. 2009). In the new design, positive ions produced from outgassing materials are no longer accelerated towards the conversion surface. The collimator is now grounded on IMAP-Lo, which has the added benefit of allowing the ceramic insulators needed for IBEX-Lo to be eliminated. This saves space and reduces risk of a high-voltage failure. The outer electrode is set to a negative potential (nominally, -3.5 kV), which repels photo-electrons in the spacecraft environment. The inner electrode is set to a positive potential (nominally, $+4$ kV), producing a dipole field across the deflectors. This dipole has a more contained potential configuration compared to the monopole field on IBEX-Lo. The modification also reduces the potential in the region surrounding the entrance system.

The IMAP-Lo deflector design was tested using the deflector EM in the UNH calibration laboratory using a high energy ion beam in vacuum (see Fig. 6). The deflection electrodes functioned properly with a maximum deflection voltage of 5 kV (proton beam energy of 2 keV). At the beginning of the test, the beam intensity was monitored with no deflection voltage applied as the system was rotated. As the system was rotated to cover the beam angular range of -10° to $+9^\circ$, the throughput fraction varied within 0.1 – 0.5% of the intensity of the undeflected beam. The count rate as a function of the deflector voltage (see Fig. 6, left) showed good agreement between the laboratory data, an analytical model for the conical deflector, and SIMION results. The test utilized an MCP imaging detector to image the ion beam passing through the collimator and being deflected by the deflection system (see Fig. 6, right). The hexagonal pattern of the collimator was observed, and the beam image moved in the direction of the outer deflector (negative voltage) as the voltage increased. This continued until the image faded when the beam reached the angular acceptance limit of the collimator. The test fully validated the deflector design.

On IBEX-Lo, a conical grid (P2) electrode was installed behind the collimator to suppress background count rates from the ion-gun effect. A major advantage of the deflection system became apparent when vibration testing showed that P2 grids just below the collimator could not survive the IMAP vibration environment during launch. These P2 grids were

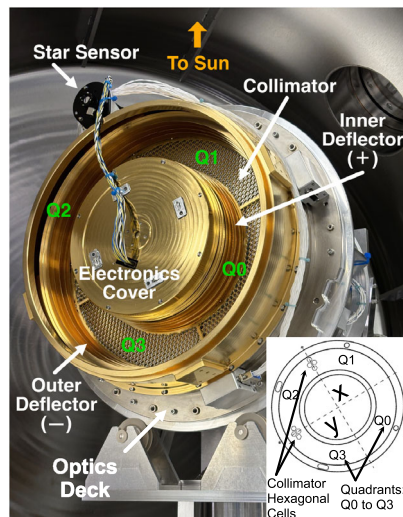


Fig. 5 Photograph of the IMAP-Lo entrance system in the Princeton calibration vacuum chamber. The star sensor is at the top-left and, in this orientation, the Sun direction is up. The collimator is divided into 4 quadrants labeled Q0 – Q3. The collimator hexagonal cells are visible. The inner deflector (at positive potential) and the outer deflector (negative potential) are labeled. The electronics cover is shown in the middle of the photograph. The optics deck of IMAP-Lo is also labeled and mounts to the IMAP-Lo calibration stage in the Princeton vacuum chamber. (Bottom-right inset) The coordinate system for the IMAP-Lo entrance system. The x- and y-axes are oriented along the collimator spokes as indicated, with the y-direction along the hexagonal vertices, and the x-direction along the hexagonal sides. The spokes divide the 4 quadrants of the collimator with quadrant 1 (Q1) containing the direction toward the Sun. The axis of pivot rotation passes between the center of quadrant 0 and quadrant 2

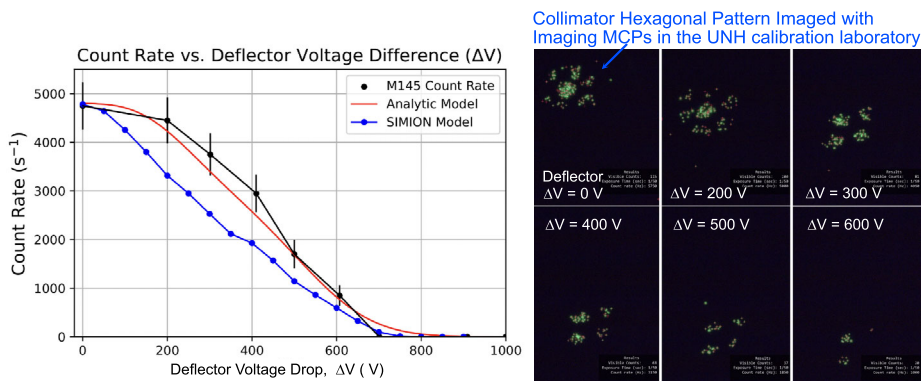


Fig. 6 (Left) Count rate vs. voltage difference between the inner and outer deflectors. The test result (black curve) is compared with an analytic model (red curve) and simulation results (blue curve). (Right) Panels show the deflected beam image revealing the hexagonal pattern of the collimator. The beam image moved lower and to the right in the direction of the outer deflector (negative voltage) as the deflector voltage increased. The voltage difference between the deflectors are indicated on each of the panels

necessary on IBEX-Lo to prevent the ion rejection system from accelerating ions formed between the collimator and the CS grid onto the CS. The deflection system on IMAP-Lo

removes the necessity for the P2 grid since an ion gun effect does not exist in the entrance system. Further, the removal of the P2 grid was shown to not substantially change IMAP-Lo electrostatic system or the energy response, and has the advantage of increasing the transmission of neutral atoms.

A potential downside of removal of the P2 grid is that ionization in the region between the CS and collimator can create a source of background. In the absence of a P2 grid, some fraction of negative ions from the CS make it back toward the collimator. Similarly, some fraction of electrons from the CS travel back toward the collimator and ionize residual gas in this region. A fraction of these ions travels back to the CS and sputter particles that masquerade as incident neutral atoms. Estimates of the effect indicate very low background rates $< 1 \times 10^{-6} \text{ s}^{-1}$, which is several decades lower than the background rates on IBEX-Lo (Wurz et al. 2009; Galli et al. 2022).

The collimator defines the fields-of-view (FOVs). The collimator design is adjusted for a broader $9^\circ \times 9^\circ$ FOV compared to the $7^\circ \times 7^\circ$ FOV on IBEX-Lo. An important lesson learned on IBEX was that it was not necessary to have high angular resolution for either the heliospheric ENAs or the ISNs at energies below 1 keV. The science requiring higher angular resolution is contained in the IMAP-Hi energy range. The increased FOV on IMAP-Lo contributes significantly to increasing the geometric factor. In addition, every quadrant uses an identical collimator, which replaces the high-resolution sector of IBEX-Lo with a nominal quadrant on IMAP-Lo, and further increases the geometric factor. These improvements in the collimator design alone increase the geometric factor of $2.3\times$ compared to IBEX-Lo.

Figure 5 shows the front of the collimator hexagon pattern. The collimator is divided into four 90° quadrants (Q0 – Q3) separated by 4 spokes at approximately $\sim 45^\circ$ (between Q1 and Q2), 135° (between Q2 and Q3), 225° (between Q3 and Q0), and 315° (between Q0 and Q1) from the vertical direction toward the Sun (note that angles increase in a counter-clockwise, right-handed sense about the outward instrument boresight direction). Optical tests of the IMAP-Lo collimator show that the FWHM FOV is $8.71^\circ \pm 0.041^\circ$, close to the designed $9^\circ \times 9^\circ$ FOV.

The IMAP-Lo collimator was designed with the same height as was used for IBEX-Lo. The angular FOV with the largest possible collection area for neutral atoms has hexagon shaped aperture holes aligned along a stack of identical photo-etched plates (see Fig. 7). The FOV is determined solely by the width $w \sim 5 \text{ mm}$ of the hexagonal holes and the total height $h \sim 27.93 \text{ mm}$ of the collimator stack extending from the entry to the exit plate. The angular width of the FOV in each direction is calculated from limiting trajectories through each aperture pair. Leakage occurs for particles that traverse into neighboring channels for tangent angles $\tan(\theta) \geq d/h$ where θ is the angle relative to the normal direction (angle from boresight) of the collimator plates and $d \sim 1 \text{ mm}$ is the line-width between hexagonal cells. The transparency T is determined by the ratio d/w , the line to the hexagon width: $T = (1 + d/w)^{-2}$, yielding a transparency of $T = 69.44\%_{-0.47\%}^{+0.33\%}$ where the uncertainties are determined from the etching tolerance ($\sim 0.01 \text{ mm}$) and the alignment tolerance ($\sim 0.01 \text{ mm}$).

Leakage of particle trajectories is eliminated using identical collimator plates that are stacked in a geometric sequence. The geometric design characteristics are the same as those used for IMAP-Hi. The largest plate separation is $h_n \leq dh/w \sim 5.6 \text{ mm}$, and the smallest separation is $h_1 \leq d/\tan \theta_{\max} \sim 0.44 \text{ mm}$, where $\theta_{\max} \leq 66^\circ$ is the largest possible incidence angle for particles. The angle $\theta_{\max}$ is limited by the precision-milled pre-collimator (with a minimum height of 2.4 mm) with trapezoid-shaped hexagon ribs that have widths slightly smaller than d (the base widths are typically $5 \mu\text{m}$ less than d). We use $h_1 \sim 0.4 \text{ mm}$, which satisfies the limit on the smallest plate separation.

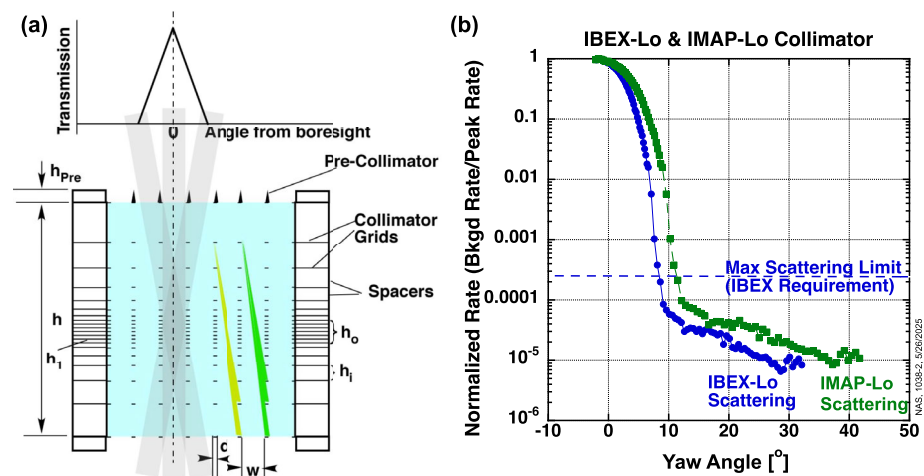


Fig. 7 (a) Aperture holes of the IMAP-Lo collimator (panel a is from Fig. 3 in Funsten et al. 2009). A fraction of particles passing through the collimator falls into the intersection of the shaded (grey) trajectory planes. The acceptance angles depend on the distance h between the plates, the opening width w , and the separation d between hexagonal cells in the collimator. The yellow and green shaded regions show the complete rejection of incident neutral particles falling outside the $9^{\circ} \times 9^{\circ}$ FOV. (b) Measured IBEX-Lo peak rate and scattering rate (blue curve) and estimated IMAP-Lo scattering rate (green curve) scaled for the increase in the full-width-half-max FWHM FOV from 7° to 9°

The geometric sequence of plate separations starts with the largest one at the exit plate and progresses to smaller and smaller separations from both the top and bottom toward the center (see Fig. 7). The 6 plates with the smallest identical separations are placed at the center of the collimator. The fraction of particles scattered at the edges of the holes into the FOV (an unavoidable effect) creates some scattering (see Fig. 7). The larger FOV of IMAP-Lo slightly increases this scattering relative to IBEX-Lo, but this source of background remains $< 0.01\%$ of the incident neutral atom rate.

Perpendicular alignment pins together with *in situ* digital microscope imaging were used to achieve precise alignment of the collimator and pre-collimator grids during the build. Optical comparator measurements were taken after the build to ensure collimator perpendicularity. Measurements of the flight model (FM) collimator resulted in an estimate for the boresight direction $X = -0.012^{\circ} \pm 0.022^{\circ}$ and $Y = 0.001^{\circ} \pm 0.038^{\circ}$ where $+X$ is directed along the spoke between Q1 and Q2, and $+Y$ is directed along the spoke between Q2 and Q3. The total geometric factor of the collimator is $2.18 \pm 0.03 \text{ cm}^2 \text{ sr}$, which accounts for collimator transmission, losses due to spokes, incomplete hexagons at the edges, shadowing at the edges, and combining the FOVs of the quadrants.

3.4 Star Sensor

Precise determination of the interstellar flow parameters relies in part on precise instantaneous directional information from the star sensor. The goal is to achieve a 0.1° pointing knowledge accuracy. The star sensor (Fig. 8) is mounted via a bracket to the grounded external housing of the outer deflector, slightly above the optics deck. Alignment was performed using an alignment cube mounted to a polished surface on the baffle housing and a second alignment cube mounted to a polished surface on the optics deck. The alignments were

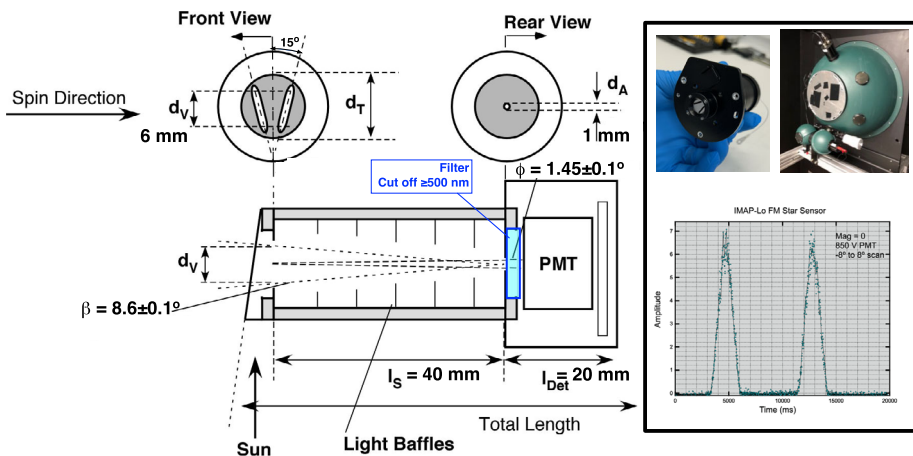


Fig. 8 The IMAP-Lo star sensor is co-mounted on a bracket to the outer deflector housing, and provides independent verification of boresight pointing during operation. (The schematic was adapted from Fig. 10 in Fuselier et al. 2009)

performed using a theodolite, showing that the star sensor is aligned with the boresight of the instrument to $< 0.05^\circ$. The star sensor's entrance aperture has two slits in the shape of a "V", a tube with light baffles for collimation, and a shortpass filter (optical density 2.0) to remove light with wavelengths above 500 nm. Beyond the shortpass filter is the photomultiplier tube (PMT). The spinning spacecraft rotates the star sensor through a star in the FOV, generating two 3° wide triangular shaped peaks. An example of these peaks from the star sensor calibration is shown in Fig. 8, bottom right. The addition of the 500 nm filter was based on a lesson learned from IBEX-Lo: the filter improves response to starlight and largely removes diffuse background from zodiacal light and the Milky Way. The 3° FWHM of the star's triangular peak is well resolved, given the timing resolution at the interface board (IFB) of 0.2 ms ($\sim 0.005^\circ$ resolution). The star sensor is sensitive to 100s of stars brighter than magnitude 3 in the visible part of the spectrum.

Star sensor signals are accumulated in each spin in 720, 0.5° bins to form histograms covering 360° . These data are transmitted to the ground and analyzed to determine the absolute star direction based on the azimuth location of the peaks in the spin plane, and the peak's time separation to determine the star's elevation.

3.5 Conversion Subsystem

Neutral atoms enter through the collimator and strike the conversion surface (CS), creating negative ions through charge conversion and recoil sputtering. In the conversion process, the CS donates electrons to incident atoms, particularly for H, D, C, and O, which have high electron affinity. Incident atoms like Ne and He have zero or very low electron affinity, respectively, and do not readily accept donated electrons to become negative ions. Since all atoms arrive at the CS with a substantial kinetic energy, they cause recoil sputtering of atoms and molecules from species present on the CS, i.e., H and O from H_2O always present on the surface, and C from the CS itself (e.g., Taglaier 1990; Vicanek and Urbassek 1990). A fraction of sputtered ions leaving the CS are negatively charged, and while molecules are typically broken up into atomic constituents, measurements from IMAP-Lo calibration

campaigns show small anomalous tracks in the TOF spectra of negatively charged molecules that can survive up to the passage through the carbon foils.

Charge conversion surfaces have been used in major planetary and heliophysics missions for decades (e.g., Wurz et al. 1997; Wurz 2000; Scheer et al. 2006; Wurz et al. 2006; Kazama et al. 2007; Wahlström et al. 2008), including the Interstellar Boundary Explorer (IBEX; McComas et al. 2009), the predecessor of IMAP for heliospheric ENA measurements. The use of conversion surfaces to measure neutral atoms was applied in instruments at missions such as IMAGE (Burch 2000), Ulysses (Witte et al. 1992), Jupiter ICy moons Explorer (JUICE) (Grasset et al. 2013), Mars and Venus Express (Barabash et al. 2006, 2007), Chandrayaan-1 (Bhandari 2005), BepiColombo (Milillo et al. 2010), and Tianwen-1 (Ye et al. 2017; Wan et al. 2020).

Charge conversion surfaces are used in the detection of ENAs to convert neutral atoms to ions, particularly for applications involving detection of low energy neutral atoms. In the energy range 10 eV to ~ 500 eV neutrals do not have sufficient energy to be detected using carbon foils directly, and conversion surfaces have to be used (Wurz 2000).

Conversion of neutrals to negative ions enables electrostatic acceleration, thereby increasing the detection efficiency, while also deflecting negative ions out of the paths of UV photons. This technique, which is typically applied below about 1 keV, was first suggested for space applications by Gruntman (1993) and was first used for the LENA instrument on IMAGE (Ghielmetti et al. 1994; Wurz et al. 1995).

A tetrahedral amorphous carbon (ta-C) conversion surface (Wieser et al. 2005), referred to commonly as a diamond-like carbon (DLC) surface, was successfully used on IBEX-Lo and is used on IMAP-Lo. The surfaces and their conversion efficiencies have been shown to be stable over periods of many years (Scheer et al. 2005, 2006). The IBEX-Lo CS has been stable for more than 15 years.

The DLC surfaces of IBEX-Lo were manufactured by Lockheed Martin Advanced Technology Center, Sandia National Laboratories, and the University of Arizona. The process for creating the DLC-coated surfaces on silicon substrates was recreated at Southwest Research Institute for IMAP-Lo. This effort enabled production of ultra smooth surfaces with surface roughnesses $R_q \leq 5$ Å and DLC film thicknesses < 50 nm (details in Sokół et al. 2024, and references therein). The surface roughness affects the angular scatter of the reflected beam and is characterized by the use of atomic force microscopy (AFM, Allenbach et al. 2018; Riedo et al. 2012; Gasser et al. 2021).

Three-inch silicon wafers (Virginia Semiconductor, Boron-doped, primary flatness 1 mm, 5 Å roughness) were cut into trapezoidal facets ~ 64 mm long and 30 mm wide at the center. The conversion surface mounts were slightly increased in size relative to IBEX-Lo to accommodate the 9° FOV. The IMAP-Lo facets were also slightly enlarged (lengthened by 0.5% and widened by 2%) to maintain coverage for the broader FOV. The modified mount combined with the increased size of the CS facets and the increased FOV helped produce a $\sim 10\%$ increase in CS yield. The edges of facets were chamfered so that they fit together, forming an annular cone (Fig. 9) inclined at 15° from the nominal incident direction of the neutrals.

The silicon substrates were precision cleaned and then subjected to Plasma Immersion Ion Deposition (PIID) to create the DLC film. The surfaces were then hydrogen terminated with a hydrogen beam in vacuum to chemically terminate exposed carbon bonding chains on the surface. Hydrogen termination removes oxygen atoms from the surface, making the surface almost inert, and lowers the surface work function without adding surface roughness (Scheer et al. 2005; Wieser et al. 2005). Measurements of the coating showed that a 46.7 ± 0.8 nm thick DLC film was deposited on each trapezoidal facet. The DLC layer had surface smoothness of 3.4 ± 0.2 Å across each sample area.

Fig. 9 Photograph of the IMAP-Lo CS facets. A total of twenty eight facets are mounted conically within the conversion subsystem

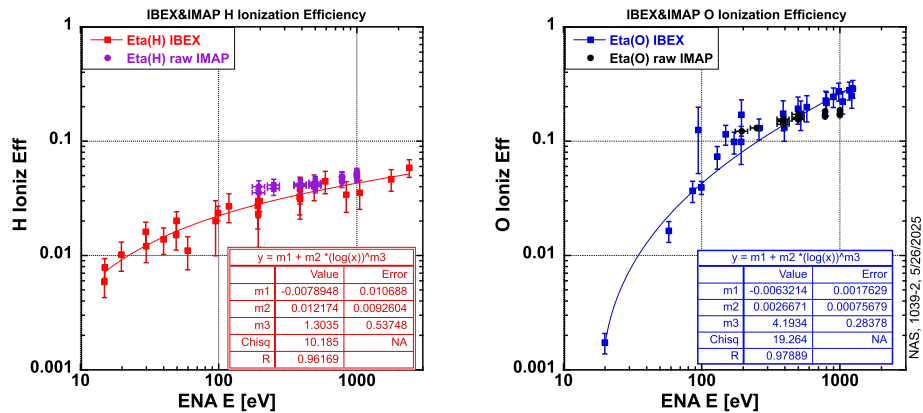
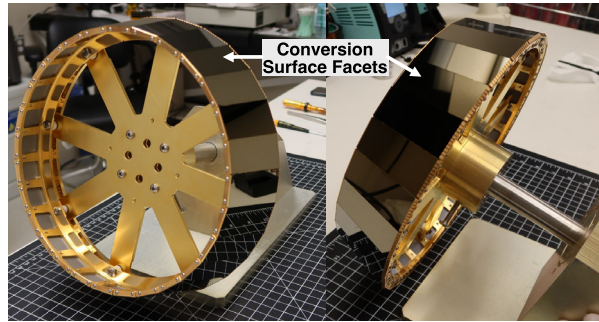


Fig. 10 Measured neutral to negative ion ionization efficiencies (or yield) of IMAP-Lo conversion surfaces for H (left panel) and O (right panel). Shown are the combined results of CS sample testing for IBEX-Lo (H red squares and O blue squares) from (Wieser et al. 2007) and for all IMAP-Lo CS samples (H purple circles and O black circles) from (Sokół et al. 2024). The empirical fits to the results by Wieser et al. (2007) over the full energy range of IBEX-Lo are also shown

The 28 CS facets are inert, have low electrical conductivity, and have high negative ion yield for impacting neutral atoms at grazing incidence ($\sim 15^\circ$). The Imager for Low Energetic Neutral Atoms (ILENA) facility (Wahlström et al. 2013) at the University of Bern was used to test the conversion surfaces. During the tests, a ~ 1 mm beam of positive ions accelerated to 90 - 1500 eV/q was directed at a CS sample at an 8° grazing incidence angle (82° to the normal to the surface), which is the default testing configuration in the ILENA facility.

Particles reflected off the CS sample were measured with an MCP imaging detector to determine the angular scatter distributions. The negative ion yield (Fig. 10) was computed based on the measured rates of neutral atoms and negative ions scattered off of each CS sample (see Gasser et al. 2021; Sokół et al. 2024). The measured negative ion yield was $2\% \pm 0.5\%$ for H, $9\% \pm 0.5\%$ for O at 200 eV and $13.5\% \pm 1.5\%$ for O at 1 keV. Sputtering yields were 1.5% to 3% for He and 3% to 5% for Ne.

The reflection efficiency, which is also energy and mass dependent, strongly affects the conversion efficiency for DLC surfaces. The surface roughness determines the fraction of negative ions that scatter specularly (mirror-like reflection). The CS smoothness at the atomic level maximizes the fraction of scattered negative ions. However, the number of par-

ticles that scatter away from the specular direction increases with energy and with angle of incidence. This result is due to the incoming particles penetrating deeper into the surface and thereby broadening the angular distribution of outgoing ions. Therefore, we find a reduction in the reflection efficiency with increasing particle energy (Wahlström et al. 2008).

Ionization and reflection efficiencies also depend on incident particle species. The O electron affinity is higher than that of H, and the O negative ion yield is $\sim 4\times$ to $6\times$ larger than that of H. The reflection efficiency also depends on mass, with O having a lower reflection efficiency than H. These differences in reflection efficiency and similar differences in energy loss between O and H are tied to the lower mass of H.

3.6 Electrostatic Analyzer (ESA)

The CS facets are held at a negative potential, causing electrostatic repulsion of negative ions leaving the CS. The ESA consists of toroidal electrostatic potential surfaces with two electrodes near the ESA entrance for focusing and accelerating negative ions into the ESA, and a third electrode near the ESA exit to guide the ions into the TOF analyzer.

The electrostatic analyzer is in the shape of a Bundt baking pan (Moestue 1973) and is almost identical to that of IBEX-Lo. The plate separation between the inner and outer ESA is large, enabling the large energy passband. UV background is suppressed with sharp fins on the outer ESA toroidal surface (see the ESA in Fig. 3). The inner ESA shell is serrated, and both the inner and outer shells were blackened to further reduce UV reflection and absorb particles on non-nominal trajectories. There are at least 3-bounces required for UV to reach the entrance to the TOF, which greatly reduced the UV background.

The energy passband $\Delta E/E$, the ratio of the FWHM ΔE of the ESA, divided by the central energy of the passband, is determined by the instrument ion optics through voltages on the CS, the P9 electrode on the opposite side of the ESA entrance from the CS, the inner ESA voltage (P5) and the P10 electrode at the exit of the ESA. The voltage levels are controlled by the U_+ and U_- set points using two precision resistor strings (Sect. 3.10), an identical configuration to IBEX-Lo. The ion optics near the ESA exit deflect the azimuthal trajectories so that the ions arrive at the TOF entrance nearly normal to the plane of the carbon foils. Ions arriving with large azimuthal trajectories may exit the ESA at large angles to the normal direction and are then lost in the TOF system by triggering only a start signal from the first foil, or triggering no start signal at all. The entire TOF subsystem is floated at 12 kV post-acceleration (PAC) to focus negatively charged particle distributions and accelerate them into the TOF subsystem.

High throughput of the ESA depends on the smoothness of the CS. If the surface is too rough, then trajectories leaving the CS at angles greater than 10° from specular reflection either strike the inner or outer ESA, or exit the ESA at relatively large oblique angles with respect to the foil-plane normal direction and therefore trigger no valid TOF signal. Tests conducted during the development of IBEX-Lo indicated that required smoothness should be $< 5 \text{ \AA RMS}$.

Incident neutral atoms produce copious numbers of electrons from the CS, particularly since UV photons also impact the CS. As on IBEX-Lo, a concentric circles of permanent disc magnets made with SmCo (Samarium Cobalt) Grade 18 materials are used to deflect secondary electrons from entering the ESA: one concentric circle of magnets on the inner (P3) and one on the outer (P9) electrodes are assembled near the entrance to the ESA (Wieser et al. 2007). An important consideration in the design of IMAP-Lo was the strength of these magnets. The disc magnets have a field strength of $\sim 0.9 \text{ mT}$ at 1.26 cm . The collection of these magnets assembled in concentric rings results in far greater field strengths than

individual disc magnets. The field strength in the gap between the P3 and P9 electrodes varies from a low of 3 mT near the midpoint of the gap to > 6 mT at the edges of the gap, exceeding the 1.5 mT field strengths required for the design with a significant margin.

Electrons generated from neutrals hitting a surface, and photo-electrons released by UV photons, are associated with an internal background identified early in the development of IMAP-Lo by studying IBEX-Lo flight data. The electron signal was largely eliminated by cutting out events with very short TOF values. However, a related internal ion background decreased the signal-to-background ratio, particularly below 200 eV: at ESA steps 4 to 1 (~ 110 eV to 15 eV) and 7 kV PAC, the internal IBEX-Lo background rate was found to vary from 0.0036 ± 0.001 to 0.0068 ± 0.001 triple-coincidence counts-per-s, increasing at each successively lower ESA step. Galli et al. (2014) and Galli et al. (2022) found representative IBEX-Lo background rates of 0.01 s^{-1} at 16 kV PAC, and 0.007 s^{-1} for a low PAC of 7 kV. The investigation into the background involved flight observations of IBEX-Lo, including those in the background mode. SIMION simulations supported the source as electrons observed in the TOF subsystem from the P10 electrode just outside the ESA exit as the background source. The P10 electrode is most strongly exposed to particles at the outer edge of the open window that is covered by the P10 equipotential grid. Ions impacting the exposed edge produce large numbers of secondary electrons and some H^- that are then accelerated into the TOF system due to the PAC potential.

A modified P10 electrode (the outside edge of the P10 window was retracted by 1 mm) was produced from a spare IBEX-Lo part to directly compare the original to the new configuration during two consecutive runs in a thermal vacuum test with the IBEX-Lo/IMAP-Lo engineering test unit. The tests demonstrated that the background was reduced by up to a factor of 4 at the lowest voltages with the modified P10 electrode configuration. Further, for the low energy ENA background we observed the background rate drops by roughly one decade over a range of temperatures from 30°C (typical IBEX temperature) to 0°C (IMAP-Lo temperatures drop into the range from -20° to -10°C). The modified design of the P10 electrode was adopted for IMAP-Lo.

IMAP-Lo temperatures are expected to be much lower than IBEX-Lo (temperature reduction by more than 20°), ensuring that IMAP-Lo background rates should be much lower.

Ion-optical simulations using the instrument response model (Sect. 5.3) were performed to study the trajectory distribution of ions scattered off the CS and the subsequent ESA throughput and TOF efficiencies. These simulations made use of observed conversion efficiencies and angular distributions from the CS for incident particles at grazing incidence angles (e.g., Wahlström et al. 2008; Sokół et al. 2024).

These simulations made use of observed conversion efficiencies and angular distributions from the CS for incident particles at grazing incidence angles. The results verified that a large fraction of ions is lost in the ESA due to the broad angular and energy distribution emitted from CS. In HiThr mode, the ESA focuses a broader angular distribution from the CS into the TOF by raising the voltage difference between U_+ and U_- for specifically tuned values.

The elimination of the highest IBEX-Lo energy step (1.8 keV) for IMAP-Lo made it possible to implement the new HiThr mode in addition to the HiRes mode of IBEX-Lo. The HiThr mode was directly tested using the IMAP-Lo Engineering Model (EM, Sect. 4) and repeated for the FM (Fig. 11). These results were roughly consistent with simulations, showing throughput increased by a factor of 3 – 4 and a broader ESA passband ($\Delta E/E \sim 1$) in HiThr mode as compared to the HiRes mode ($\Delta E/E \sim 0.7$). There are important differences between the simulations and the calibration curves, which are currently under study.

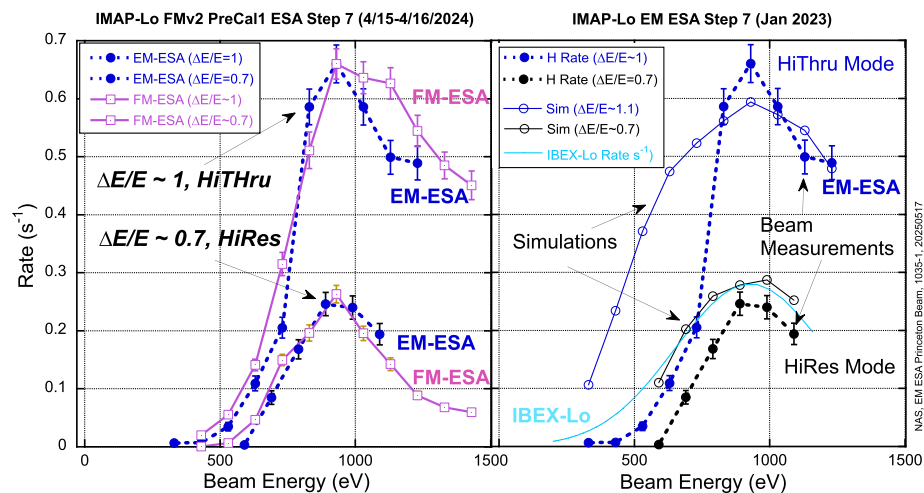


Fig. 11 (Left) The IMAP-Lo EM and FM electrostatic analyzer tests in ESA step 7 for HiThru and HiRes modes. These tests demonstrated the successful implementation of increased electrostatic throughput in HiThru mode. (Right) EM testing of the ESA, TOF, and CS at Princeton in Jan 2023 showed that the HiThru mode increased the instrument throughput. The observed response was narrower than simulated (solid curves) or inferred from IBEX-Lo (light blue solid curve)

3.7 Time-of-Flight (TOF) Subsystem

The TOF subsystem uses two semi-circular rings of carbon foils (1 $\mu\text{g}/\text{cm}^2$) and a microchannel plate (MCP) near the exit of the TOF section as a triple coincidence time-of-flight spectrometer for negative ions that traverse the foils and stop at the MCP (see Figs. 12 and 13). The TOF system distinguishes between H, D, C, and O negative ions. The use of triple coincidence measurements suppresses random background events. Distinguishing between the sputtering products and the products from electron capture for H^- , D^- , C^- , O^- depends on the branching ratios observed in TOF spectra for H, D, C, O. Note that interstellar He and Ne are only registered by their sputtered products. An example of TOF spectra are shown in Fig. 12 during a test in which a 400 eV Ne beam sputtered H^- , C^- , and O^- from molecules on the CS (e.g., H_2O , and OH, and C from a hydrocarbon layer on the CS and from the CS lattice itself). Note the large differences between HiRes and HiThru modes. The increase in total counts shows the increased throughput of HiThru mode, and the increased fraction of H results from better collection of negative H ions scattered over a broad angular and energy range from the CS. H ions are scattered more broadly from the CS due to the lower mass of these atoms (particularly compared to C). The application of HiThru mode enabled collecting larger fractions of negative ions emitted over broadened angular and energy distributions.

Negative ions are accelerated into the entry foil ring by the +12 kV PAC voltage, critical for accelerating and simultaneously focusing ion trajectories into the TOF. The +12 kV PAC voltage provides sufficient energy for negative ions to traverse the entry foils, the exit foils, and retain sufficient energy to be registered by the MCP with relatively high efficiencies for full triple coincidence. Typical H and O TOF detection efficiencies are $\sim 90\%$ for double coincidence, and $\sim 36\%$ for triple coincidence.

A negative ion coming from the ESA traverses the first entrance foil and releases secondary electrons that are electrostatically accelerated and steered to the outer portion of the

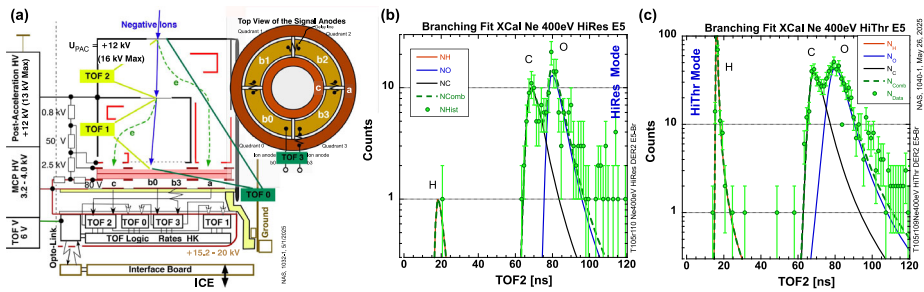


Fig. 12 (a) Schematic of the TOF mass spectrometer. The TOF is rotationally symmetric about the left hand side of the figure. A negative ion (blue arrow) coming from the ESA traverses the first entrance foil, and knocks off secondary electrons (green dashed curve) that are electrostatically accelerated and steered to the outer portion of the microchannel plate (MCP). The MCP signal from the ion (labeled “a” on the anode below the pink MCP stack) is the first “start-a” signal. The particle subsequently traverses the second, exit foil. Secondary electrons from this exit foil are steered to the inner edge of the MCP stack and create the “start-c” signal. The particle finally strikes the MCP and creates “stop-b0” and “stop-b3” signals via the delay lines in the signal anode (top right inset). The signal anode is divided into four sectors (b0, b1, b2, and b3) with delay lines between three sectors, and a fourth delay line between b3 and the ion anode. (panel a adapted from Fig. 1 in Möbius et al. 2008). (b, c) TOF spectra from the IMAP-Lo flight instrument taken during X-Cal at Los Alamos National Laboratory (LANL). A 400 eV neutral Ne beam was used in this test. The masses observed include H, C, and O sputtered from the CS and from the breakup of water (and OH, and other hydroxyl group molecules) on the CS. The test was conducted with ESA step 5 in HiRes mode at (b, center energy of 200 eV, passband 135 eV–265 eV), and HiThr mode (c, center energy 213 eV, passband 107 eV–320 eV)

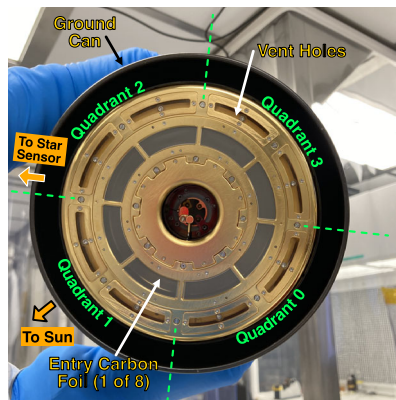


Fig. 13 Photograph of the front entrance of the TOF mass spectrometer in the Princeton test vacuum chamber. The 8 ultra-thin ($1 \mu\text{g}/\text{cm}^2$) carbon entry foils are visible and are mounted on high transmission grids (333 lines-per-inch). Vent holes surround the perimeter of the Time-of-Flight section, and following IBEX-Lo heritage, minimize possible acoustic damage. The entire TOF section is elevated to 12 kV, which accelerates negative ions so that they pass through the entry foils, the exit foils and strike the micro-channel plate

microchannel plate (MCP), creating the first “start-a” signal. The particle subsequently traverses the second exit foil and secondary electrons are steered to the inner edge of the MCP stack creating the “start-c” signal. The particle finally strikes the MCP and creates “stop-b0” and “stop-b3” signals via the delay lines in the signal anode (Fig. 12a, top right inset). By analyzing the signal delay between anode b0 and b3, the quadrant for the stop signal

is determined. Using the TOF_2 time delay between start-a and start-c, the TOF_1 time delay between start-c and b3, the TOF_0 time delay between start-a and b0, and the TOF_3 time delay between b0 and b3, we determine all TOFs of the particle through the TOF system. The TOFs combined with the energy-per-charge from the PAC voltage determines the mass and therefore species of incident ions. The sectoring of the signal (from the b-anode) provides an important additional means for background rejection and can be used to estimate internal background rates.

The major differences in the TOF system between IMAP-Lo and IBEX-Lo are: (1) A SHAPAL insulator was used on IMAP-Lo as the main insulator since it simultaneously acts as an electrical HV isolator and a good thermal conductor (<https://precision-ceramics.com/materials/shapal/>); (2) The PAC voltage was reduced from 16 kV on IBEX-Lo to 12 kV on IMAP-Lo to reduce the risk of discharge. This change was made after EM testing showed evidence of partial discharge above 13 kV; (3) IMAP-Lo uses thinner $1 \mu\text{g}/\text{cm}^2$ foils as compared to $2 \mu\text{g}/\text{cm}^2$ foils used on IBEX-Lo for improved TOF resolution; (4) Spokes within the TOF subsystem were thickened by 1 mm since early vibration testing showed that these spokes were vulnerable to vibration failure.

There was a high-voltage partial failure on IBEX-Lo well after the IBEX prime mission. This partial failure was the result of a long eclipse that put extreme thermal stress on the main MACOR insulator within the TOF unit, likely causing cracks in the MACOR and compromising the HV isolation. This motivated the use of SHAPAL for the insulator on IMAP-Lo. SHAPAL acts as a thermal conductor rather than a thermal insulator. A trade study during the IMAP-Lo design phase successfully qualified the SHAPAL insulator.

3.8 TOF Board

The TOF electronics board (see TOF electronics in Fig. 3) is located within the TOF subsystem behind the MCPs and the anode board. The TOF board monitors individual rates associated with the two start and two stop signals, measures the timing between the signals, performs logic timing determinations, and produces event measurements. The TOF board also gathers housekeeping information related to the MCP and TOF system.

The TOF electronics are operated as a set of linear voltage differentials: the PAC defines the largest voltage of the system ~ 12 kV, the MCP voltage of $\sim 3.2 - 4$ kV is stacked on the PAC voltage, which provides the acceleration of the secondary electrons to the MCPs, the MCP bias, and a small voltage to the signal anode (Fig. 12a). The TOF board ground is referenced to the signal board potential and powered by a floating 6 V supply. Since the entire system is at high voltage, there is a high voltage gap between the TOF optics section and the ground-referenced TOF HV supply. Signals from the TOF board are transferred to the IFB on the TOF HV electronics section through two optical links, one for signals (commands) into the TOF board, and the second for signals (telemetry) to the IFB. The IFB controls TOF and PAC high voltages and is connected to the E-Box through a serial port. The E-Box supplies conditioned low-voltage power, ± 12 V and $+5$ V, to the IFB, which in turn distributes voltage and power to the TOF HV supply.

Secondary electrons and ions deposit charge into the MCP stack, which triggers electron avalanches that are collected on the signal anodes (start-a, start-c, and stop-b0, -b1, -b2, and -b3, top right of Fig. 12). The anodes are biased to $+200$ V relative to the back of the MCP to accelerate the electrons from the MCP. On the TOF board, there are four TOF ASIC chips (Paschalidis 2002; Paschalidis et al. 2003) that combine the signals to give the ion flight times over the entire ion path from the entry foil to the stop MCP.

A valid double coincidence event requires either of two start signals and any stop signal: either TOF_0 , TOF_1 , or TOF_2 together with TOF_3 are valid double coincidence signals. A

valid triple coincidence event requires both start signals and a stop signal. A valid “golden” triple event must also meet the criterion that the TOF over the full 60 mm distance from the entry foil to the stop MCP, including the line delay ($\text{TOF}_0 + \text{TOF}_3$) is equal to the sum of TOF_2 over the 30 mm from the entry to the exit foil and TOF_1 over the 30 mm from the exit foil to the MCP. This criterion is used to define the checksum:

$$\text{checksum} = \text{TOF}_0 + \text{TOF}_3 - (\text{TOF}_2 + \text{TOF}_1), \quad (1)$$

which typically has a magnitude of less than 1 ns. This criterion reduces background due to random double coincidences, and eliminates events where any one TOF is close zero.

There were five significant changes made to the TOF board based on IBEX-Lo lessons learned: (1) additional event logic is included to remove or reduce background events. These include time-of-flight rejection thresholds (programmable and adjusted as per in-flight calibrations), and measurement veto requirements to identify invalid background events; (2) additional rate counters are included for double and triple coincidence counters, and discarded event counters; (3) ability to read the triple coincidence counters is enabled within each 6° spin-bin, rather than just once per 60° spin sector; (4) interleaving housekeeping and event telemetry is used to maintain science time resolution requirements for the direct events and rate counters; (5) capability is included to enable/disable TOF board telemetry (disabled at power-up). The rate rejection counters are critical since electron rates can overwhelm the FPGA. Because the electrons predominantly trigger specific coincidence regions (e.g., combined values of TOF_0 , TOF_1 , and TOF_2), they can be rejected before overwhelming the telemetry pathway.

3.9 TOF High Voltage Power Supply (TOF HVPS) and Data-Link

Power for the TOF subsystem is supplied by the TOF HVPS (Fig. 3) via three power supplies: one for the Post Acceleration (PAC) bias (~ 12 kV) of the carbon foils and for the steering potential (steering for secondary electrons from the foils); one for the Microchannel Plate (MCP) potential (3.2 – 4 kV); and one to power the TOF board electronics and anode (+6 V, LV6).

Control of the TOF HVPS is provided by the IFB including: 8-bit, +5 V digital-to-analog converters for TOF HVPS control of the voltage setpoints and over-current protection; +3.3 V logic signals and a pass-through from the test port of the HV disable plug; 12 bit analog-to-digital converters for analog housekeeping data from the TOF unit and the TOF supply.

The TOF HVPS supply is based on the heritage design of the IBEX-Lo TOF HVPS supply. The TOF supply is housed on a Nickel and Copper plated Ultem housing. The metal plating provides a solid ground reference and enclosure for the HV supply and provides shielding for the electromagnetic fields generated inside the supply. The plating thickness for the Ultem housing of the TOF HV Supply, as carried over from IBEX-Lo, includes a thick $125\text{ }\mu\text{m}$ (5 mil) thick Cu layer. The plating thickness is needed to provide a suitable shield with an adequate skin depth.

The TOF HV cascades are contained within the housing, and the IFB board, the PAC control board, and the MCP/LV6 board are mounted to the exterior of the housing. Three bayonet-style high voltage feed-throughs connect the TOF HVPS to the TOF optics section.

Communication between the IFB and the TOF board occurs across the HV gap between the grounded HV supply and the TOF optics at high voltage (the potential difference can be up to ~ 17 kV, which combines a maximum of 13 kV on the PAC and 4 kV on the MCPs).

Table 3 IMAP-Lo optics settings for the HiRes and HiThr modes

Energy Step	U_+ (V)	U_- (V)	Center Energy, H (eV)	Mode
1	46.84	−16.38	16.33	HiRes
2	84.29	−28.85	30.47	”
3	156.65	−53.35	55.76	”
4	302.21	−101.95	106.26	”
5	578.02	−194.16	200.04	”
6	1177.29	−395.19	404.96	”
7	2274.55	−762.78	787.29	”
1	68.12	−33.00	17.19	HiThr
2	126.85	−61.66	32.36	”
3	237.52	−115.66	59.48	”
4	461.40	−224.89	114.41	”
5	884.47	−431.33	213.70	”
6	1804.66	−880.33	437.36	”
7	3491.84	−1703.56	838.88	”

The datalink consists of two boards: a sender and receiver on the ground potential side of the gap, and a sender and receiver on the high voltage side of the gap. The latter shares its board with the high voltage distribution network (the pie board) of the TOF subsystem. The emitter and receivers consist of an IR LED and a photodiode.

3.10 Bulk High Voltage Power Supply (Bulk HVPS)

Voltages for other parts of IMAP-Lo come from the Bulk HVPS, which is mounted to the back of IMAP-Lo (Fig. 3). The HVPS consists of an HVPS controller board and an HVPS bulk board. The HVPS controller board receives commands from the C&DH board to program the HVPS bulk voltages, digitizes HVPS feedback voltages and sends them back to the C&DH board. The Bulk HVPS board supplies the deflection system voltages, the outer deflector $D_- = -3.5$ kV, the inner deflector $D_+ = 4$ kV, the optics voltages $U_+ \leq 3500$ V and $U_- \geq -1800$ V, and the star sensor PMT voltage operated at $\gtrsim -800$ V, depending on in-flight commissioning.

Ion optics voltages on electrodes are determined by the set point voltages U_+ and U_- and the high-resistance resistor strings. The E-Box controls these five high voltages and commands the TOF for a particular science or engineering mode. Set points for the optics voltages in HiRes and HiThr modes are shown in Table 3. The HiRes mode was designed to match the first seven energy channels of IBEX-Lo so that flight IBEX-Lo data can be compared with IMAP-Lo data. The energy channels were determined using the Monte Carlo response model correlated with calibration measurements.

3.11 Integrated Common Electronics (ICE), E-Box, and Other S/C Interfaces

The E-Box (Fig. 2) provides the electrical interface between IMAP-Lo and the S/C Bus. The E-Box (shown in the block diagram of IMAP-Lo in Fig. 14) comprises the following boards: the pivot control card (PCC) board that controls the pivot platform; the Command and Data Handling (C&DH) board, which transmits and receives data from the IFB, controls the TOF

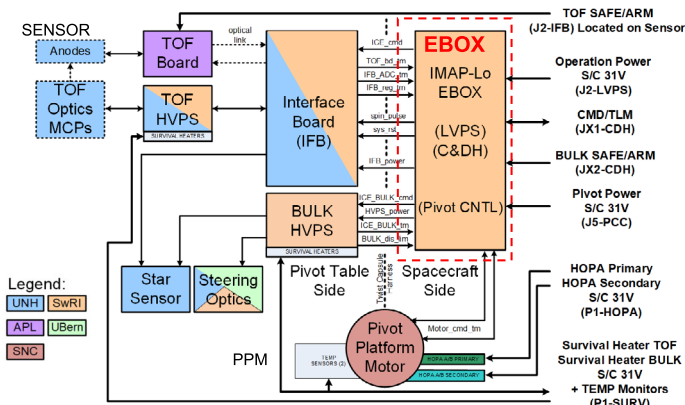


Fig. 14 IMAP-Lo block diagram showing the electronics interfaces between the E-Box, the IFB, Bulk HVPS, the PPM, the star sensor, the TOF HVPS, and the TOF subsystem. Box fill colors indicate the institutions responsible for electronics within the instrument. The E-Box contains three boards: the PCC, which controls the PPM; the low voltage power supply (LVPS); and the command and data handling (C&DH) board. Box fill colors indicate the institutions responsible for electronics within the instrument

unit through the IFB, and controls the Bulk HVPS; and the Low-Voltage Power Supply (LVPS), which receives 31 V power from the S/C, and supplies conditioned secondary power at +3.3 V, ± 12 V and +5 V to the IFB. The LVPS also powers the Bulk HVPS via ± 12 V, and +3.3 V secondary supply outputs. In addition to the interface to the E-Box, there are services from the S/C at 31 V that control the primary and secondary launch locks, and the survival heaters on the TOF HVPS and the Bulk HVPS.

The C&DH board is powered by ± 12 V, +5 V, and +3.3 V rails, and provides the data interface to the S/C, collects analog telemetry (Bulk HVPS and TOF HVPS currents and voltages, and LVPS currents and voltages, temperatures, and C&DH secondary voltages), controls the Bulk and TOF HVPS, collects data from the TOF, performs processing for these data, and provides an interface to the pivot platform controller. The C&DH board features a SPARC V8 LEON2-FT processor supported by a radiation tolerant FPGA, as well as MRAM, SDRAM, and PROM memories. The C&DH board includes a 50 MHz clock source, which the FPGA uses to generate a 25 MHz clock. The latter is used for the C&DH board's internal logic and to generate the processor's source and UART clocks. The FPGA controls general purpose I/Os, provides loopback function for board verification, performs ADC conversion for analog telemetry, generates a tick timer with 1 μ s resolution and generates various interrupts related to data acquisition, S/C communication and timers.

Since accessing MRAM and PROM memories requires different protocols, the C&DH FPGA works as an intermediary between the Processor and those memories. Further, the FPGA provides the following: (1) the S/C interface module featuring a 115.2 kbps UART, the detection of the virtual 1 pulse-per-second signal aligned with the leading edge of the first command byte, and EDAC protected FIFOs to store received S/C commands and to store data to be transmitted to the S/C; (2) the spin module, which includes a spin counter with 320 μ s resolution, and 60 spin-bins also with 320 μ s resolution; (3) the heartbeat module that provides an aliveness pulse; (4) the instrument power module that controls the power switches on the LVPS, which are turned on or off by specific code words; (5) the mission elapsed timer module that provides a coarse counter (1 sec resolution) and a 20-bit fine

Table 4 On-orbit data rates

Product	Bits-per-second	Comments
Housekeeping	24.4	976 bits every 40 s, 976 bits/s in commissioning
Direct Events	204	Maximum rate, 6 events/s with 34 bits per event
H Histogram	74	Various start, stop rates and monitor H rates accumulated over ~ 420 s packet
O Histogram	74	Various start, stop rates and monitor O rates accumulated over ~ 420 s packet
Star Sensor	384	720 samples per spin (in 0.5° bins) with 8 bits per sample
Total	760.4	Maximum bit rate, average ~ 686 bits-per-second

counter (1 μ s resolution) to timetag direct events, histogram data, housekeeping, and other data.

On orbit data products and rates processed by the E-Box are shown in Table 4. The total data rate is 760.4 bps maximum, and ~ 686 bps average. Almost half the data rate is made up of the 720 star sensor 8-bit samples collected each spin for maintaining accurate pointing determination.

4 The IMAP-Lo Engineering Model (EM)

An IMAP-Lo EM was developed early in the IMAP-Lo program. While IMAP-Lo is a heritage instrument based on IBEX-Lo, many of the subsystems had not been replicated for over a decade, and some of the parts used for IBEX-Lo had become obsolete or unavailable. This was particularly true for electronics parts such as operational amplifiers. The EM was used as a vehicle to solidify the team's understanding of where technical issues may slow development.

EM calibration verified the TOF ion optics and the properties of the ESA (see Fig. 11). The EM testing revealed the electrostatics associated with the HiRes and HiThr modes enabled comparison with instrument response simulations.

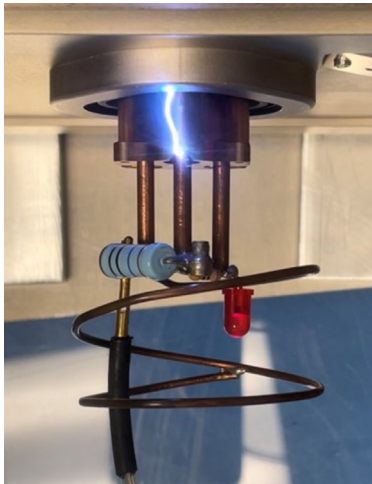
The fabrication and testing of the EM collimator for IMAP-Lo and IMAP-Hi required a development effort to achieve and demonstrate the required bore-sight alignment and perpendicularity. A dark Laboratory and an integrating sphere were used to calibrate the EM star sensor.

The EM TOF HVPS provided for critical testing of HV stability. This effort highlighted the need for robust plating of the TOF HVPS housing, particularly near the HV gap (Fig. 15).

During testing of the EM HVPS integrated with the TOF optics, start-a count rates increased to the detection limit of the TOF board (26 kHz). A series of tests indicated that the grounded TOF optics housing was picking up radiated power and resonating at high-frequencies (100's of MHz to 10's GHz range) due to electromagnetic pickup. The start-a signal was so prominent because it was received on the perimeter of the anode ring (see Fig. 12), the nearest signal region on the anode to the cylindrical TOF inner can. This start-a noise was eliminated by adding a bypass capacitor (1 nF capacitor rated to 8 kV) from the TOF inner can to the LV6 virtual ground. In the FM design, this is implemented with the bypass capacitor across the MCP supply.

Another result from EM testing was the high TOF efficiencies, which were higher than for IBEX-Lo due to the use of thinner 1 $\mu\text{g}/\text{cm}^2$ carbon foils. C and O TOF peaks were

Discharge path on EM TOF HV



FM TOF HVPS showing the feedthrough chimney

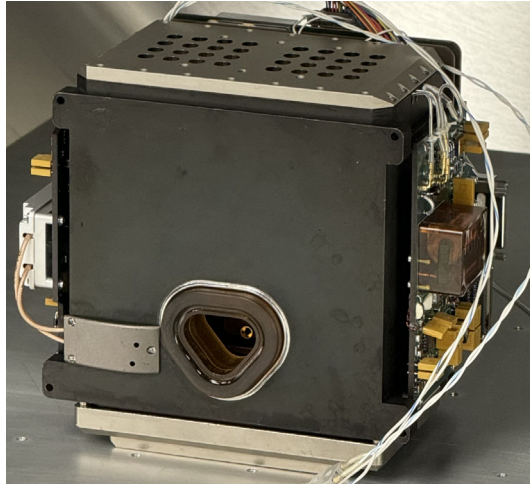


Fig. 15 IMAP-Lo EM testing on TOF HVPS revealed that the unit could discharge, particularly at voltages above 13 kV. The image on the left shows the discharge path during a test in a N_2 purged aquarium at 13.5 kV. The high-voltage feedthrough is connected to a load resistor to simulate the MCP load. The discharge path is the shortest distance to ground. The right image shows the FM TOF unit. The metal on the high voltage chimney was milled off to reduce likelihood of discharge, and a bus wire was soldered into a channel surrounding chimney to remove triple-points that can cause partial discharge or complete discharge

narrower than on IBEX-Lo and these peaks were clearly identifiable at a PAC voltage of only 7 kV, albeit with low 2% triple efficiency. At these low 7 kV PAC voltages, C and O peaks were not visible in IBEX-Lo due to the greater energy loss in the thicker $2 \mu\text{g}/\text{cm}^2$ foils. The results confirm that $1 \mu\text{g}/\text{cm}^2$ foils improve TOF efficiencies and allow for better identification of species from the CS.

5 Calibration and Cross-Calibration

The IMAP-Lo instrument was calibrated at the Princeton Space Physics Laboratory (Sect. 5.1) throughout the year in 2024 and cross-calibrated at the Los Alamos National Laboratory (LANL) beam facility (Sect. 5.2) in February 2025. Calibration results are consistent with the instrument response model (see details in Sect. 5.3) at energies of 55–1500 eV. Calibration of the lowest two energy steps relies more heavily on the response model, together with low energy measurements from the University of Bern, to determine energy loss on the CS. Results verified higher geometric factors than observed on IBEX-Lo, in part due to enhanced TOF efficiencies from the ultrathin $1 \mu\text{g}/\text{cm}^2$ foils.

Geometric factors derived from the response model are folded into the simplified response function. This approximation of instrument response is the convolution of the point-spread function associated with incident neutrals and the transmission function through the instrument (Schwadron et al. 2009).

5.1 Flight Instrument Calibration

IMAP-Lo was assembled and tested in four stages: TOF calibration, pre-cal 1, pre-cal 2, and final calibration. First, the TOF analysis subsystem was tested using an ion beam at

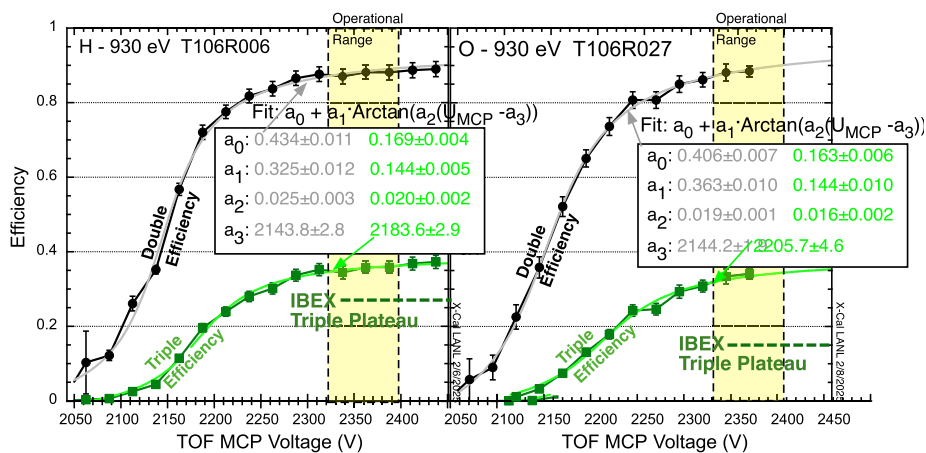


Fig. 16 TOF efficiency curves as a function of the bias voltage across the MCP stack for tests T106R006 (930 eV H) (left panel) and T106R027 (930 eV O) (right panel). Shown are the triple (green curve) and any double (black curve) efficiencies. The yellow shaded region shows the operational range targeted for flight operations. Coefficients for the fit equation, $\eta = a_0 + a_1 \arctan[a_2(U_{\text{MCP}} - a_3)]$, are shown in grey for double-coincidence events and in green for triple-coincidence events

Princeton. The initial tests were done to identify the TOFs of various ion species in the TOF subsystem.

The absolute TOF efficiencies and TOF characteristics (TOF_0 , TOF_1 , TOF_2 , and TOF_3 as a function of species and beam energy) are determined independently from the rest of the instrument. Singles rates and all double coincidence rates are monitored in the TOF and the triple coincidence rate is determined from individual direct events (DEs). Each direct event is associated with the TOF values: TOF_0 , TOF_1 , TOF_2 , and TOF_3 . Double coincidences correspond to any two combinations of TOF_0 , TOF_1 , and TOF_2 . Silver triple events correspond to any event that triggers all four TOFs. Golden triple events, in addition, satisfy the checksum criterion.

The IMAP-Lo TOF subsystem is fully self-calibrating because triple and double coincidence rates can be used to derive detector efficiencies, in calibration or in flight (Funsten et al. 2005). Figure 16 shows TOF double and triple coincidence efficiencies for H and O as a function of MCP voltage. These figures also include the IBEX-Lo triple efficiency plateaus (green dashed lines). For both H and O, the observed triple efficiencies are significantly larger than those observed by IBEX-Lo. These tests were performed during X-Cal with a 930 eV H beam (left panel) and 930 eV O beam (right panel).

Following TOF calibration was pre-cal 1 where the instrument was partially assembled and an aperture mask was placed over the entrance system to allow only a small portion of the beam to reach the center of one CS facet. These initial pre-cal 1 tests verified basic instrument performance in a configuration similar to the EM tests. In particular, the results revealed the energy passbands in HiThr and HiRes modes, and initial assessments were made for the instrument geometric factors. Pre-cal 2 followed pre-cal 1 with a fully integrated sensor to prove performance and functionality before environmental testing.

During instrument calibrations (pre-cal 1, pre-cal 2, and final calibration) we primarily utilized neutral beams (H from 30 eV to 1.9 keV, and O from 450 eV to 930 eV). Figure 17 shows the installation of the fully assembled IMAP-Lo instrument into the calibration vacuum chamber for the final calibration tests. The instrument was installed on a rotation stage

Fig. 17 Image of the IMAP-Lo FM installed in Princeton Space Physics Calibration System for final calibration

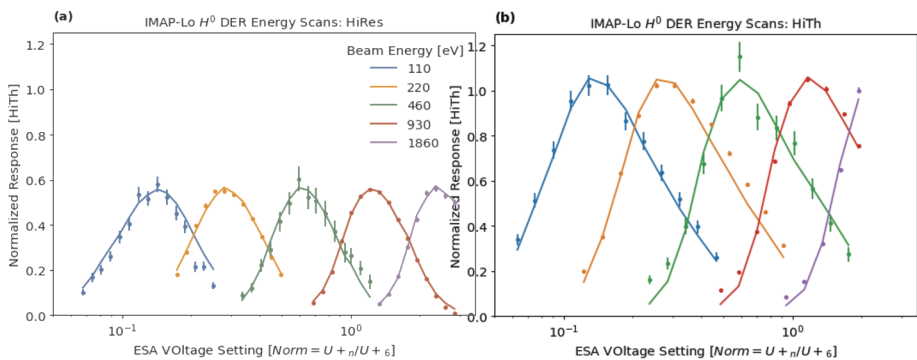
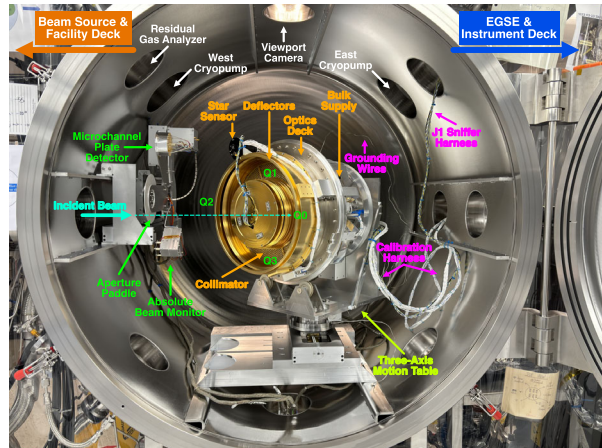


Fig. 18 Calibration differential energy response curves compared to the response model. Shown (a) are the response curves in HiRes mode, and (b) the response curves in HiTh mode. In both bases, the beam consists of H atoms, and the ESA U_+ and U_- voltages were varied for a fixed beam energy. The peaks in each case correspond to the U_+ and U_- ESA settings for ESA steps 3 to 7. The x-axes are U_+ / U_{+6} where U_{+6} is the positive ESA voltage at ESA step 6. The y-axes are the response functions normalized to the average HiTh mode rate of 1.0. The absolute magnitudes of each peak are determined by the instrument geometric factors at each ESA step

mounted to a 2-axis motion table. The rotation stage allowed tests of the azimuth response, while the motion table was used to test the radial and yaw-angle response. In Fig. 17, the neutral beam source is at the left. Details about the Princeton neutral beam generated via a charge-exchange vessel are provided in Rankin et al. (2025).

Figure 18 shows sample results from the IMAP-Lo final calibration. The five peaks correspond to IMAP-Lo energy bins 3 to 7 and show the energy response. These data were obtained by scanning the energy acceptance over the beam energy. The measured fluxes were normalized in the Figure for comparison between ESA steps and ESA modes. The derived $\Delta E/E$ is roughly constant over the energy range of each ESA step.

Figure 19 shows the energy loss of H and O from the CS as a function of incident neutral energy. Since voltages on the ESA were designed to pass negative ions with 15% less than the incident neutral energy, a larger energy loss from the CS translates into a higher incident neutral energy.

Fig. 19 Fractional energy loss (fraction of energy lost from the incident beam energy after CS interactions, causing sputtering and ion conversion) on the IMAP-Lo CS using a pencil beam for neutral atoms. The energy loss was determined from the peak of the ESA passband for recoil ions compared to the input energy of the neutral beam

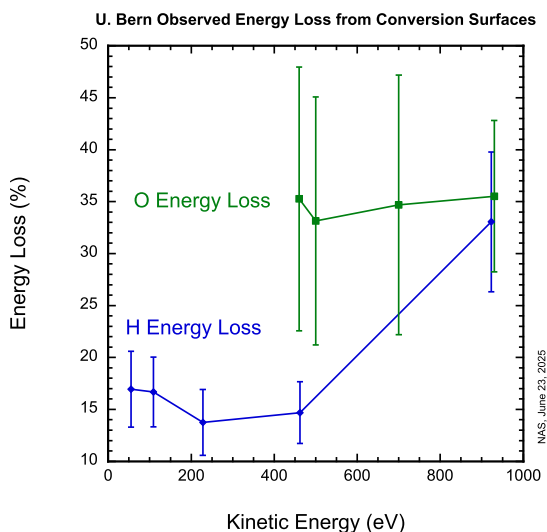


Figure 12 shows a TOF spectrum measured by IMAP-Lo. A 400 eV Ne neutral beam was directed into the IMAP-Lo instrument. The instrument was set to detect neutrals centered at 135–265 eV in HiRes mode (left), and 107–320 eV in HiThr mode. These energies are slightly lower than the beam energy since atoms sputtered by Ne are distributed at lower energies than the incident beam. The H, C, and O TOF peaks are caused by recoil sputtering of negative ions from the CS by the neutral Ne beam. There is no peak identified as Ne produced by true conversion of neutral Ne. Unlike species of neutral atoms with high electron affinity, Neon’s electron affinity is nearly zero, and it does not form a stable negative ion that could be registered by the TOF system. For other beams with atoms that have high electron affinities (i.e., H or O), most of the signal observed near beam energies originates from conversion to a negative ion.

A larger fraction of H, and wider TOF peaks for C and O indicate that HiThr mode collects broader angular and energy distributions of sputtered ions from the CS. Notably, H atoms are emitted over an increased angular range from the CS in comparison to C since H has much less mass than the C atoms that make up the CS.

Figure 20 provides another result characterizing the effects of sputtering from the conversion surface. The plot compiles runs from final calibration with 400 eV and 700 eV Ne beams in HiRes mode. The x-axis shows the central ESA energy divided by the beam energy and the y-axis shows the C/O ratio derived during a given run. The CS surface is largely composed of C, and a water layer exists on the CS. Therefore, both C and O are expected sputter products. The fraction of C relative to O depends in part on the work function associated with sputter yields. Based on IBEX-Lo, the characteristics of sputtering should remain throughout the life of the mission, providing diagnostics of the incident ISN He distribution (Schwadron et al. 2022).

During calibration of IMAP-Lo, it was essential to establish the uniformity of the yield and throughput as functions of azimuthal angle and position of the neutral beam on the CS. Figure 21b shows results of “snake scans” where a beam was rastered across the entrance system, providing an effective image of the CS negative ion yield and throughput. While some variation associated with spokes and hexagon cell structure are expected in both azimuthal and snake scans, the response is not strongly dependent on position. Figure 21a

Fig. 20 C/O ratios as a function of ESA central energy divided by beam energy obtained in the final calibration runs with 400 eV and 700 eV Ne beams. A polynomial fit is provided to characterize the sputtering response

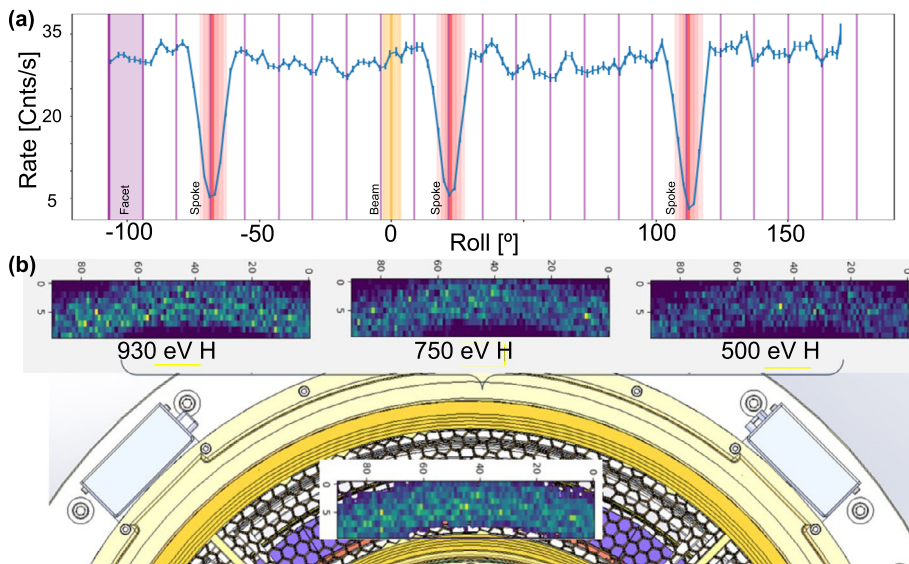
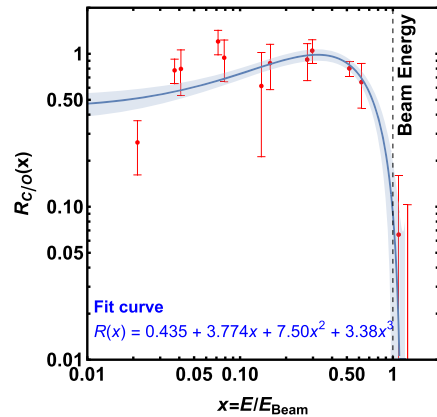


Fig. 21 (a) Final Cal Roll Response, where 3 spokes separate quadrant 3 (Q3), Q0, Q1, and Q2 from left to right. This roll response was performed with the neutral beam collimated by a 14 mm collimator paddle. (b) Sweep scans (snake scans) result averaging 930 eV, 750 eV, 500 eV neutral H beams for precise beam alignment to center of IMAP-Lo CS facet at Quadrant 0 (12 o'clock in this figure)

shows the azimuthal or roll response of the instrument measured during final calibration runs. There is some axial variation in collection efficiency across the collimator spokes, with about an $\sim 80\%$ reduction observed between the spoke region and the quadrant center. The quadrant-to-quadrant collection efficiency appears relatively uniform. Spatial dependence introduces a low overall 4% root-mean-square (RMS) variation, which slightly increases G uncertainties.

Table 5 shows geometric factors for each energy step for triple coincidence H. The energy transmission is approximated as a Gaussian, $T(E) = (2\pi\sigma)^{-1} \exp[-(E - E_p)^2/(2\sigma^2)]$, where E_p refers the neutral peak energy, and σ is the root-variance that characterizes the energy spread. These factors were determined from the calibration and include $\Delta E/E$, colli-

Table 5 IMAP-Lo H geometric factors

Energy Step	Neutral Peak Energy (eV)	Root-variance σ (eV)	Energy Range ¹ (eV)	$G \times 10^5$ (cm ² -sr-eV/eV)	Mode
1	16.33 ± 0.28	4.58 ± 0.11	10.9 - 21.7	7.0 ± 4.9	HiRes
2	30.47 ± 0.43	8.55 ± 0.15	20.4 - 40.5	7.9 ± 5.5	"
3	55.76 ± 0.89	16.1 ± 0.34	36.8 - 74.7	9.7 ± 6.8	"
4	106.3 ± 1.9	29.4 ± 0.71	71.6 - 140.9	11.2 ± 3.0	"
5	200.0 ± 3.4	55.2 ± 1.3	135.0 - 265.0	14.0 ± 4.5	"
6	405.0 ± 7.3	115.5 ± 2.9	269.0 - 541.0	17.7 ± 2.0	"
7	787.3 ± 22.0	240.0 ± 7.0	504.7 - 1069.9	22.5 ± 1.4	"
1	17.19 ± 0.22	7.53 ± 0.17	8.3 - 26.1	16.6 ± 10.1	HiThr
2	32.36 ± 0.36	13.7 ± 0.3	16.2 - 48.5	19.2 ± 11.7	"
3	59.48 ± 0.77	24.3 ± 0.5	30.9 - 88.1	22.7 ± 13.8	"
4	114.4 ± 1.3	45.5 ± 0.8	60.8 - 168.0	27.3 ± 9.8	"
5	213.7 ± 3.0	90.0 ± 2.1	107.7 - 319.7	33.8 ± 9.5	"
6	437.4 ± 5.3	187.1 ± 3.7	217.1 - 657.7	44.4 ± 4.9	"
7	838.9 ± 11.7	361.1 ± 8.3	413.7 - 1264.0	56.9 ± 2.9	"

¹Neutral atom energy range from lower to upper half-maximum

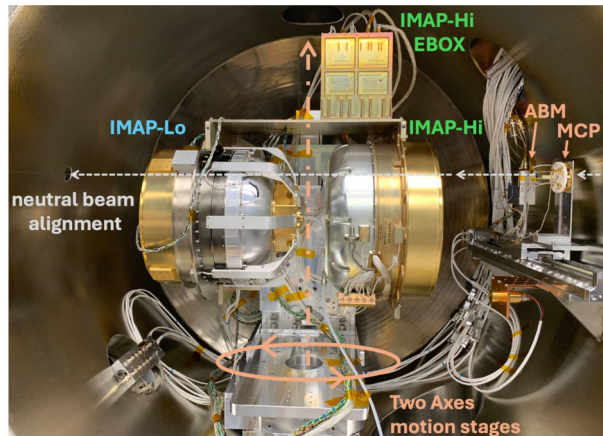
mator solid angle FOV, all of the efficiencies of transmission through the collimator, internal grids transmission, effects of the spokes that separate each azimuthal quadrant, the energy dependent conversion efficiency, and TOF efficiencies.

Wurz et al. (2009) discussed IBEX background sources. Numerous background sources are from positive ions produced at or behind the collimator exit that are accelerated to high energies by the collimator positive voltage. On IMAP-Lo, the use of an ion deflector (Sect. 3.3) greatly reduces the background sources. However, even with deflection, electrons and ions can collide with atoms making up the residual gas within the instrument. As a result, the background flux from the collimator depends on the residual gas pressure in this part of the instrument. To reduce this gas pressure, the instrument has significant vent paths that bypass critical regions, electronics are vented separately from optics, there are no vent paths to the spacecraft interior, and materials in the optics path were carefully chosen for their low outgassing properties. For pressures $< 5 \times 10^{-7}$ Torr within the instrument, background rates remained extremely low ($< 6 \times 10^{-4} \text{ s}^{-1}$ at ESA step 7). The in-flight environment will have more than a decade lower pressure than could be achieved in the laboratory, which reduces the background rate even further.

5.2 Cross-Calibration (X-Cal)

The IMAP-Lo FM was cross-calibrated with the IMAP-Hi 45 FM at LANL in February 2025. The primary objectives of this X-Cal campaign were: (i) to verify the full functionality of the IMAP-Lo FM following spacecraft environmental testing (after acoustic and vibration tests, but prior to spacecraft thermal vacuum testing), and (ii) to compare the beam fluxes measured by both the IMAP-Lo and IMAP-Hi instruments given the same neutral beam and chamber conditions. The comparison was conducted under the requirement of achieving a relative beam flux agreement better than 10% between the two instruments and an overall absolute beam flux accuracy within 25%, based on the derived geometric factors. The “same

Fig. 22 Chamber configuration of IMAP-Lo and IMAP-Hi 45 X-Cal at LANL. The ABM and imaging MCP are shown. Two motion stages were implemented to rotate the two instruments in and out of the neutral beam



beam” was defined as having less than 10% variation in beam intensity, as measured by the ABM before and after each X-Cal activity, a level of stability expected to be under 5% based on LANL’s experience. Each instrument was required to complete a scan at normal incidence, collecting statistically meaningful data with less than 5% uncertainty.

The X-Cal measurements were performed using neutral H beams at 500 eV, 750 eV, and 930 eV. The IMAP-Lo FM was calibrated without its flight PPM (not needed for X-Cal) or flight E-Box, which both remained on the S/C. During the X-Cal campaign, the incident beam was aligned to illuminate a single CS facet on IMAP-Lo through its collimator, targeting the center of the facet located in Quadrant 0 - the same facet previously calibrated during final calibration at the Princeton Space Physics Laboratory. A carbon-foil ABM and an imaging MCP detector were used throughout the campaign to monitor beam intensity and spatial uniformity.

The X-Cal campaign included a series of differential energy response measurements across multiple neutral species and energies. The objective was to supplement existing calibration data and refine the final instrument calibration parameters.

Figure 22 shows a photograph of the LANL vacuum chamber setup, with both IMAP-Lo and IMAP-Hi 45 FMs mounted on a large crane. During the X-Cal campaign, the IMAP-Hi E-Box operated inside the vacuum chamber, while the IMAP-Lo EM E-Box remained outside the chamber in a cleanroom environment. A narrow, pencil-like neutral beam was aligned to sequentially target specific locations on both instruments. The ABM was routinely employed before and after each critical measurement to monitor beam stability and provide the reference absolute beam flux. Beam alignment was controlled via a two-axis motion system (vertical and rotational), allowing precise positioning of each instrument in the beam path.

Figure 23 summarizes the IMAP-Lo and IMAP-Hi 45 X-Cal results. The left panel displays the measured beam rates (solid curves), inferred from the incident flux recorded by both instruments. These rates account for each instrument’s geometric factor and response function. The dashed lines represent the mean count rates from the ABM, recorded before and after each beam measurement. For IMAP-Lo, the 500 eV and 930 eV beams align with its highest two ESA steps 6 and 7, respectively. In contrast, for IMAP-Hi, the 500 eV and 750 eV beams correspond to its lowest two ESA steps 1 and 2, respectively. The right panel of Fig. 23 shows the inferred beam fluxes normalized to the ABM reference rates. These results confirm that the relative flux accuracy is within $\pm 10\%$, and the absolute flux accuracy is within the mission requirement of $\pm 25\%$.

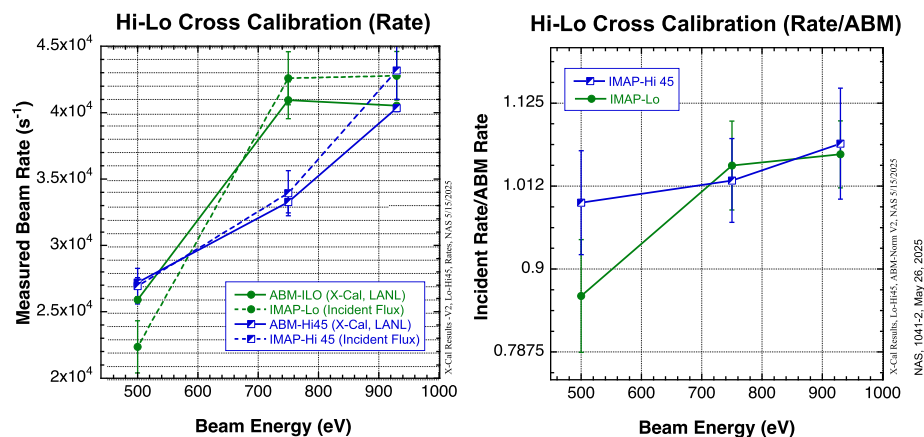


Fig. 23 IMAP-Lo versus IMAP-Hi cross calibration compared to the reference ABM beam rate. (Left) measured ABM beam rates compared to rates inferred by IMAP-Lo and IMAP-Hi. (Right) IMAP-Lo and IMAP-Hi rates normalized by ABM rates

5.3 IMAP-Lo Response Model

The IMAP-Lo response model, a Monte Carlo simulation used to characterize the instrument, has become an essential tool in calibration. The model has been refined based on calibration measurements, and measurements of the CS, enabling instrument behavior to be characterized over low energy ranges (< 50 eV) where only limited calibration data exist. The response model complements the calibration measurements, helps determine incoming beam characteristics, and enables the determination of neutral atom sources at the core IMAP-Lo science.

The energy response of the ESA-CS subsystem is characterized using a two-stage Monte Carlo simulation based on neutral particles incident on the conversion surface. In the first stage, CS calibration data are used to model surface-scattering interactions. The second stage simulates the resulting ion trajectories as they pass through the optical path of the ESA. Figure 24 is a schematic diagram of the IMAP-Lo response model. The model was first trained by IBEX-Lo data, then assimilated with the IMAP-Lo calibration dataset.

In stage one of the response model, the distribution of particles incident on the CS is defined as shown in Fig. 24 labels 1 to 3. Surface interactions - including energy loss, energy dispersion, scattering recoil angular distributions, ionization efficiency, sputtering probability - are modeled using a Monte Carlo routine.

Calibration measurements for O and Ne show a linear relationship between velocity perpendicular to the conversion surface and the angular recoil scattering FWHM (Neuland et al. 2014). Using this relationship, each particle's incident velocity is used to define a unique scattering probability distribution which is then sampled to give a distribution of ions recoiling from the CS. The conversion efficiency, defined as a function of incident velocity (Wahlström et al. 2008), is incorporated as a weight factor for each particle in the final integration.

In the second stage (Fig. 24 labels 4 and 5), the trajectories of recoil ions through the electrostatic fields are fully simulated in 3D using SIMION, assuming axial symmetry in the IMAP-Lo electrode geometry. Recoil ions are tracked from their origin at the CS, through the ion optics, into the ESA, and ultimately to the first TOF entrance foil. The surviving

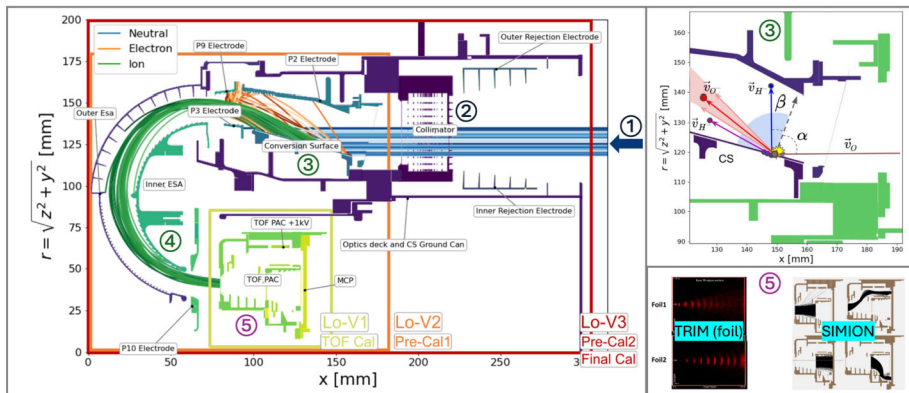


Fig. 24 Cutaway drawing of the IMAP-Lo instrument subsystems overlaid with SIMION ray-tracing. Neutral atoms enter the instrument from the right in this geometry, they pass between the two rejection electrodes, which sweep away charged particles, and pass through the collimator with 9° FWHM FOV. The neutral atoms then strike and reflect off the conversion surface, where a fraction of them convert to negative ions. The specularly reflected negative ions are pre-accelerated into a toroidal ESA, and filtered according to energy-per-charge, while electrons are removed by a ring of magnets behind the P9 pusher electrode. After passing through the ESA, ions are post-accelerated into the TOF subsystem, where they pass through two consecutive $1\text{-}\mu\text{g}/\text{cm}^2$ C foils that produce secondary ions which are guided to the outer/inner annular start sections of the chevron configuration microchannel plate (MCP). The ions continue to the central stop section of the MCP. The stop measurement combined with the two start measurements produces three independent TOF measurements

population of incident neutral particles that produce detectable recoil ions is then integrated, yielding a survival probability as a function of incident energy.

The final stage of the simulation provides the efficiencies, energy losses and TOFs through the IMAP-Lo TOF subsystem based on TRIM simulations. This final stage is often not needed to characterize differential energy responses, particularly since efficiencies through the TOF are self-calibrated both in calibration and in-flight.

6 Operations and IMAP-Lo Data Products

IMAP-Lo operations (Sect. 6.1) are planned to optimize scientific output balancing interstellar neutral atom and ENA mapping objectives. IBEX-Lo has relied heavily on accurate data products often in the form of maps for both ISN rates and ENA fluxes. These lessons from IBEX-Lo are applied to streamline the production of IMAP-Lo data products (Sect. 6.2).

6.1 IMAP-Lo Operations

IMAP-Lo operations are designed to be simple, repetitive, and straightforward. Spacecraft repointing maneuvers are executed daily, and immediately following these, PPM pointing operations are executed. The commands used to point the PPM are uploaded on a monthly basis and included in the monthly planning process for instrument and spacecraft activities.

PPM pointing is limited to the range 75° - 105° for the first year of operations with the platform going back to 90° every other day for the creation of complete 1-year heliospheric maps.

Table 6 IMAP-Lo Data Products

Product	Level	Comments
Star Sensor Data	1B	Processed star sensor data
Annotated DEs	1B	
Histograms & Monitor Rates	1B	
Pointing Sets	1C	
S/C Frame Flux Maps	2	
S/C Sputter & Bootstrap Corr. Maps	2	
Helio Frame Kin. Maps	2	
Helio Frame CG Maps	2	
H, O Rate Maps	2	
Surv. Prob. Corrected Maps	3	Fluxes projected to 80 au

In science mode, the instrument is set at a fixed energy step for two spins, so that the entire energy range is sampled in 14 spins. Data products are collected over a complete acquisition cycle consisting of two sweeps through the 7 ESA steps, through 28 spins total.

Periodically, IMAP-Lo will move to an alternative energy stepping mode based on the particle populations it observes. The stepping tables are loaded to MRAM and can be activated via either real-time or time-tag commands.

6.2 Data Products and Algorithms

IMAP-Lo forms data products and telemetry in the same format, regardless of science mode. Table 6 shows the data products that are accumulated at various levels. At level 1B, IMAP-Lo data are accumulated over 28 spins (about 7 min), whereas star sensor data are accumulated in 720 bins for each spin.

Pointing sets represent the basis for all IMAP-Lo maps. These data are composed of DEs and exposure time accumulated in 3600×40 arrays. Each element represents a $0.1^\circ \times 0.1^\circ$ binned direction on the sky in the S/C frame. Within each pointing (roughly 1 day of science operations) an average spin direction is defined. The azimuthal angle is the horizontal angle about the spin direction and is binned in 3600 azimuthal angle elements. The central elevation angle of the pointing set is the average angle between the instrument boresight and the nominal spin direction. Off-angles (or deviation angles) in elevation are defined relative to the central elevation angle. These off-angles are binned within 40 elevation angle elements, allowing for variation of the boresight up to 2° either toward or away from the central elevation angle. The off-angle binning allows for coning, nutation, and other angular variations that inevitably affect the instantaneous pointing of the instrument. For example, on IBEX-Lo, pointing variations are typically observed within 0.2° .

The IMAP-Lo mapping algorithms represent generalized approaches to creating ENA flux maps and rate maps based on the pointing sets created for level 2 data products. Products presented here include both level 2 maps and level 3 survival-probability corrected maps.

IMAP-Lo observes both ENAs and ISN atoms by measuring H^- counts and O^- counts. H ENAs and ISN He and H will produce mostly H^- by direct charge-exchange or surface sputtering, whereas ISN O produces predominantly O^- and with some contribution of H^- , particularly in HiThr mode. ISN Ne produces a mix of O^- and C^- via sputtering with some H^- contribution.

IMAP-Lo ENA map products represent detected H and O atoms:

- **Energetic Neutral Atom Maps.** ENAs (H) enter the instrument, undergo charge-exchange on the conversion surface to create H^+ . ENAs are characterized by differential energy flux and are subject to ionization loss over their long trajectories through the heliosphere. ENAs move along geodesics, much like light. Therefore, ENA flux maps correspond to the intensity from the incident ENA directions within the heliosphere. ENA H maps also have contributions from ISN He at ESA step 4 and below.
- **Oxygen Atom Maps.** Oxygen atom rate maps are predominantly interstellar in origin, although additional sources of O cannot be ruled out, and ISN Ne contributes sputtered O^+ . Since O has an even higher electron affinity than H, there is significant conversion on IMAP-Lo's CS that generates O^+ from incident O atoms. The peak in O-maps is near the incidence direction of O ISN flow, which depends on the ecliptic longitude and the PPM angle at the time the map was created. The maps are comprised of events classified based on O^+ TOF signatures with contributions of C^+ ions sputtered from the CS.

For each source of neutral atoms, there is also a background. Based on results from IBEX-Lo, the dominant background in the instrument emanates from surfaces (such as the P10 electrode). While IMAP-Lo eliminated or reduced these sources of background, it is likely that some low-level background will be observed. Because the background results from particles that pass through the ESA, it is important to treat the background as a separate source of particles. Therefore, differential fluxes are computed without subtracting backgrounds. The background fluxes are calculated separately and carried with IMAP-Lo data products.

Interstellar Neutral Atoms (ISN) are mapped, characterized, and analyzed as rates since they often include contributions from multiple species. There are only two fundamental rate products delivered:

- Light Atom maps associated predominantly with ISN H and He;
- Heavy Atom maps associated predominantly with ISN O and Ne.

IMAP-Lo ENA maps are in the ram hemisphere (look directions with a component in ram direction) since the ram directions naturally suppress background. The associated products include the following:

- **Maps in the S/C frame with no corrections.** The basic map or slice of data that is useful for science includes differential fluxes of H atoms in the S/C reference frame without any corrections applied.
- **Maps in the S/C frame with sputter and bootstrap corrections.** The sputter-bootstrap corrected map or slice of data includes differential fluxes of H atoms in the S/C reference frame with corrections for sputtering by O and Ne, and the bootstrap correction for sputtering by higher energy ENAs.
- **Maps in the inertial frame with kinematic corrections.** The kinematic inertial map or slice of data from differential fluxes of H atoms in the heliospheric reference frame includes (1) sputtering and bootstrap corrections; (2) translation to the heliospheric reference frame via a kinematic transform from the S/C frame. The energy of ENAs in this reference frame varies over the map since the S/C frame energies are translated directly into heliospheric frame energies based on a frame of reference transformation between the moving S/C frame and the heliospheric inertial reference frame.
- **Compton-Getting Maps in the heliospheric inertial frame.** The CG-corrected map or slice of data includes differential fluxes of H atoms in the heliospheric inertial reference frame with sputtering and bootstrap corrections, and the Compton-Getting (CG) transform. These products are placed in the heliospheric reference with the CG transformation

applied, and the energy of ENAs in this reference frame is constant over the map. The CG correction interpolates the energy of ENAs in the S/C frame so that the ENA energy is constant in the inertial reference frame. The CG correction can be applied reliably at ESA steps 5 - 7 (200 eV and above). At lower ESA steps 1-4, the ENA energy shift between reference frames is larger than the ESA bandpass, and larger than the difference between the next higher ESA level and the ESA level analyzed. At these low energies, the energy interpolation is done over large portions of each energy step, and the CG corrections introduce substantial variations, particularly where the background approaches the measured intensity. The CG correction is also only applied to products taken in HiRes mode since HiThr mode opens the energy acceptance making the transformation to a fixed energy in the heliospheric frame non-linear.

Maps are created at 6° resolution, below the 9° field-of-view (FOV) of IMAP-Lo. Each of the maps are corrected for survival probability at level 3.

7 Summary

IMAP-Lo is a single pixel, large geometric factor neutral atom imager mounted on a pivot platform designed to take the next major leap in mapping low energy ENAs from the boundaries of the heliosphere and determine the precise flow properties and composition of the local interstellar medium. The imager benefited from almost 20 years of development, test, and on-orbit operation of the IBEX-Lo sensor. The lessons learned from IBEX have had enormous impacts on the IMAP-Lo design, development, and testing.

IMAP-Lo detects 5 eV to 1 keV heliospheric neutral H, D, He, O, and Ne in 7 broad energy bands in multiple energy resolutions. The instrument uses charge conversion on a diamond-like carbon surface to convert neutrals into negative ions. The negative ions are then accelerated and deflected through IMAP-Lo's electrostatic analyzer and into the time-of-flight subsystem, which is floated to 12 kV. The high potential of the time-of-flight subsystem post-accelerates negative ions to ~ 12 keV so that they have sufficient energy to pass through two $1 \mu\text{g}/\text{cm}^2$ foils and trigger the microchannel plate detector. The time-of-flight subsystem uses a triple coincidence from the two foils and the microchannel plates to distinguish species: H, D, He, O, and Ne from the heliosphere and from the local interstellar medium.

IMAP-Lo is mounted on a pivot platform, which enables articulation of the IMAP-Lo boresight to detect interstellar neutral atoms throughout much of the year while simultaneously mapping the boundaries of our heliosphere in ENAs. The ability to observe the local ISN flow from multiple vantage points and orientations remove fundamental measurement ambiguities that have thwarted absolute determinations of local interstellar flow properties.

IMAP-Lo has undergone a rigorous campaign of testing: first with the IMAP-Lo engineering model, then flight model calibration, and finally cross-calibration with the IMAP-Hi flight model. These tests demonstrate that IMAP-Lo fully meets the requirements for the IMAP mission. The IMAP-Lo geometric factors exceed $6\times$ those of IBEX-Lo, a result that is almost 50% higher than expected based on the design. Improvements in time-of-flight efficiencies through the use of ultrathin foils and other opportunistic improvements enabled significant increases in collection power while also reducing the risk of high-voltage discharge during flight operations. The increased geometric factor on IMAP-Lo coupled with the ability to pivot the boresight open an enormous array of new scientific opportunities that contribute to IMAP's quantum leap in understanding the composition and properties of the local interstellar medium, the evolution of the outer heliospheric boundaries, and the physics of interstellar-heliospheric interactions.

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Declarations

Competing Interests The authors have no conflicts of interest to declare that are relevant to the content of this article.

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Authors and Affiliations

N.A. Schwadron^{1,2}  · M.M. Shen² · C. Amaya³ · S. Begley⁴ · J. Bower¹ · J. Brody³ · M. Bzowski⁵ · G. Cardarelli¹ · E.R. Christian⁶ · G. Clark⁴ · C. Collura⁴ · C. Cook⁴ · S. Cortinas³ · R. Crosier³ · M. Cully⁴ · A. De Los Santos³ · J. Dominico⁷ · N. Dutton⁴ · S. Ellis¹ · K. Fairchild¹ · J. Ford³ · C. Frost¹ · H.O. Funsten⁸ · S.A. Fuselier^{3,9} · A. Galli¹⁰ · M. Gargano¹⁰ · C.E. Garza³ · J. Gasser³ · M. Gkioulidou⁴ · M. Granoff¹ · N. Hall¹ · J. Hanley³ · D. Heirtzler¹ · C. Hersman⁴ · E. Hertzberg³ · S. Homan³ · I. Householder¹ · S. Hoyt¹¹ · H. Islam¹ · P. Janzen¹² · T. Jones¹ · N. Kerman¹¹ · B. King¹ · E. Klages⁷ · M.A. Kubiak⁵ · G. Lafferty¹¹ · W. Lam⁴ · C. Lomax⁴ · S. Longworth¹³ · K. Mathews⁴ · D.J. McComas² · C. McLeod¹⁴ · E. Möbius¹ · K. Nelson⁴ · A. Neto de Mattos¹³ · J. Niehof¹ · C. Nunez³ · D. Piazza¹⁰ · S. Pope³ · F. Rahmanifard¹ · D. Reisenfeld⁸ · J. Reiter¹¹ ·

C. Reno² · S. Rizo-Patron³ · H. Rodriguez³ · J. Rushforth¹ · C. Schiferl³ · C. Schlemm⁴ · M. Segler³ · J.M. Sokół³ · N. Souitat¹³ · D. Suffern⁷ · P. Swaczyna⁵ · M. Tapley³ · K. Taylor³ · J. Teifert² · W. Toczynski³ · P. Twombly¹ · S. Vogler¹ · C. Weaver¹ · A. Wester¹³ · M. Wieder⁴ · E. Wilber¹ · P. Wilson³ · R. Winslow¹ · P. Wurz¹⁰

✉ N.A. Schwadron
nschwadron@unh.edu

- ¹ Institute for the study of Earth, Oceans and Space Science, Space Science Center, University of New Hampshire, 8 College Road, Durham, 03824, NH, USA
- ² Department of Astrophysical Sciences, Princeton University, 171 Broadmead, Princeton, 08544, NJ, USA
- ³ Southwest Research Institute, 6220 Culebra Rd, San Antonio, 78238, TX, USA
- ⁴ Applied Physics Laboratory, Johns Hopkins University, 11101 Johns Hopkins Rd, Laurel, 20723, MD, USA
- ⁵ Space Research Centre PAS, CBK PAN, Bartycka 18 A, Warsaw, 00-716, Poland
- ⁶ NASA Goddard Space Flight Center (GSFC), 8800 Greenbelt Rd, Greenbelt, MD, 20771, USA
- ⁷ Sierra Space, Broomfield, CO, USA
- ⁸ Los Alamos National Lab. (LANL), Los Alamos, NM, USA
- ⁹ Department of Physics and Astronomy, University of Texas at San Antonio, San Antonio, 78249, TX, USA
- ¹⁰ Physics Institute, Space Research and Planetary Sciences, University of Bern, 3012 Bern, Switzerland
- ¹¹ Laboratory for Atmospheric and Space Physics, University of Colorado, Boulder, CO, USA
- ¹² The New Mexico Consortium, Los Alamos, NM, USA
- ¹³ Earth Ocean and Space Science, Southwest Research Institute, 8 College Road, Durham, 03824, NH, USA
- ¹⁴ Space Systems Integration, LLC, PO Box 220427, Chantilly, 20153, VA, USA